Chair of Software Modeling and Verification (Computer Science 2) | RWTH AACHEN UNIVERSITY

Bachelor Thesis at the
Software Modeling and Verification Group (MOVES)

Expected Visiting Times in Interval Markov Chains

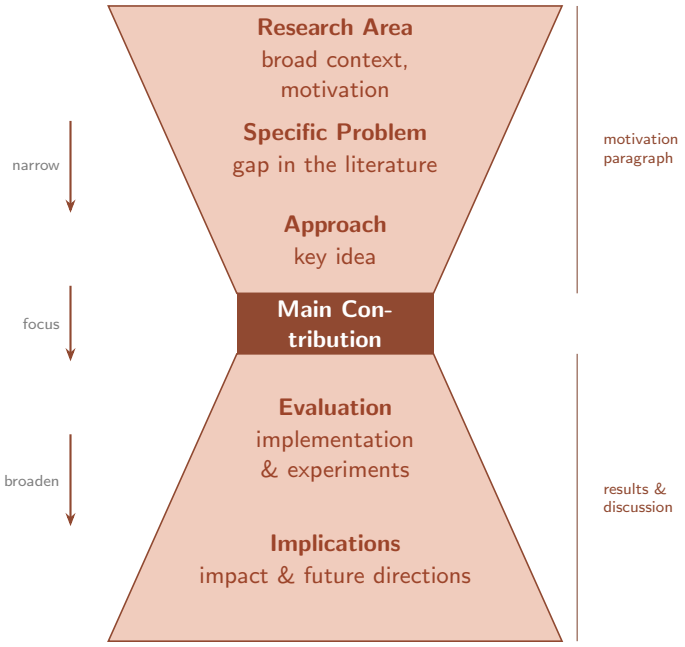
by
Jesper Nilsson

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First examiner: Joost-Pieter Katoen
Second examiner: insert text
Thesis advisor: Hannah Mertens

Abstract



Hourglass structure of a good abstract. Start broad (research area and problem), funnel down to the specific contribution, then widen again to state results and implications.

Write a self-contained summary of **roughly 150–250 words** following the hourglass structure above. Cover the following five points in order:

1. **Research area.** Name the field (e.g., strategy synthesis, probabilistic program verification, operational semantics) and state in one sentence what the field studies and why it matters.
2. **Specific problem.** State precisely which problem this thesis addresses and why it is challenging or not yet fully solved by existing approaches.
3. **Approach.** Describe the key idea of your contribution in one or two sentences. Avoid mathematical notation and abbreviations, and use easily understandable language.
4. **Theoretical results.** Name the main property proven (soundness, completeness, an equivalence result, or a complexity bound) and state it informally.

5. **Implementation and evaluation (if applicable).** Name the tool and the benchmark set and give the headline result: how many more examples are solved compared to the baseline and what is the runtime overhead.
6. **Implications.** Explain how your results solved the issue or filled the gap you pointed out beforehand.

Acknowledgements

I want to express my most profound appreciation to my supervisor!

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Chapter 1

Introduction

The introduction motivates the topic and provides an overview of the thesis. It can be seen as a longer version of the abstract that follows the hourglass structure again, but with emphasis on the top part of the hourglass. However, you can now use citations and explain everything in more detail. It should be accessible to any computer scientist without prior knowledge of the specific area.

The background and motivation part should cover the following points:

1. Name the research area and describe informally what objects are studied (e.g., markov automata, probabilistic programs, POMDPs) and what property is of interest (e.g., reachability, complexity, safety).

2. Explain why the property matters.

3. Give a small, self-contained example that illustrates the problem informally.

4. Explain why answering the question automatically is non-trivial and what makes the problem hard in general.

The open-problem part should contain:

1. A summary of what is already known in the area (prior results, existing automated techniques, tool support).

2. A precise identification of the gap: e.g., no automated procedure exists, existing techniques are unsound or incomplete for a relevant subclass, or no efficient algorithm is known.

3. A clear statement of what this thesis aims to achieve.

Related Work

Discuss the most relevant prior work (either here, or in a seperate chapter depending on the scope of existing work). For each cited work, address the following:

1. What does the work do? Describe the approach in one to three sentences and do not just list references.
2. How does it relate to this thesis? Does it solve a special case, use a different technique, or target a different formalism?
3. What limitation motivates your own contribution?

Contribution

List one item per concrete contribution. Each item should:

1. Identify what is new (a technique, a criterion, a framework, an implementation, or an experimental evaluation).
2. State the property proved, if applicable (soundness, completeness, both, or a complexity bound).
3. Reference the theorem, or section in which the contribution appears.

Structure

End the introduction by outlining the remaining chapters. Give at least one sentence per chapter describing its content. Do not simply write "Chapter X presents the main results."

Example: "The remainder of this thesis is organized as follows. Chapter 2 introduces the formal background required throughout this thesis, including ... Then, **??** presents ... **??** describes the implementation and the experimental evaluation. Finally, Chapter 6 summarizes the results and discusses directions for future work."

Chapter 2

Preliminaries

2.1 Notation

We denote the entry in the i-th row and j-th column of a matrix $\mathcal{T}$, as $(\mathcal{T})_{ij}$. We will use the notation $\underline{n} = \{1, 2, \ldots, n\}$ with $n \in \mathbb{N}$ for sets. For vectors v, we denote the i-th entry as $v(i)$ Cardinality of set S as $|S|$. $\bigcirc$ $\diamond$

$\exists!$ to denote: precisely one.

We use *Iverson Brackets* $[\![B]\!]$ to denote the function that maps a Boolean expression B to 1 or 0 such that

$$[\![B]\!] = \begin{cases} 1 & \text{if B is true,} \\ 0 & \text{otherwise.} \end{cases}$$

2.2 Matheteil

Linear Order Measurable set (used in def. 3.12)

Definition 2.1 (Convergence of Geometric Series)
The geometric series $\sum_{k=0}^{\infty} q^k$ converges for all $q \in \mathbb{R}$ with $|q| < 1$, where

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}\,, \quad \text{for } |q| < 1$$

Definition 2.2 (Gershgorin circle theorem)
Let $\mathcal{T} \in \mathbb{C}^{n \times n}$ be a square complex matrix, then for every eigenvalue λ of $\mathcal{T}$ it is

true that

$$|\lambda - (\mathcal{T})_{ii}| \leq \sum_{j=1 \wedge i \neq j}^{n} |(\mathcal{T})_{ij}| \ ,$$

for any $i \in \underline{n}$ and where $|\cdot|$ denotes the absolute value, i.e. magnitude.
In other words, all eigenvalues of $\mathcal{T}$ lie on a disk in the complex plane centered at $(\mathcal{T})_{ii}$ with radius $\sum_{j=1 \wedge i \neq j}^{n} |(\mathcal{T})_{ij}|$.

Matrix invertible if 1 not eigenvalue

Definition 2.3 (Row-Stochastic Matrix)
A square matrix $\mathcal{T} \in \mathbb{R}_{\geq 0}^{n \times n}$ is row-stochastic, if and only if $\sum_{j=0}^{n} (\mathcal{T})_{ij} = 1$ for all rows $i \in \underline{n}$.

When considering each row as a function $f_i(j) = (\mathcal{T})_{ij}$, these functions are discrete probability distributions.

Definition 2.4 ((Strictly) Row-Substochastic Matrix)
A square matrix $\mathcal{T} \in \mathbb{R}_{\geq 0}^{n \times n}$ is row-substochastic, if and only if $\sum_{j=0}^{n} (\mathcal{T})_{ij} \leq 1$ for all rows $i \in \underline{n}$.
$\mathcal{T}$ is strictly row-substochastic if and only if ther exists at least one row i' with $\sum_{j=0}^{n} (\mathcal{T})_{i'j} < 1$

Theorem 2.5
If a matrix $\mathcal{T}$ is strictly row-substochastic, then it has no eigenvalue $|\lambda| \geq 1$.

Proof 1
Let us assume that $|\lambda| > 1$ is an eigenvalue of $\mathcal{T}$. As $(\mathcal{T})_{ij} \in \mathbb{R}_{\geq 0}$, using def. 2.2

$$|\lambda - (\mathcal{T})_{ii}| = |\lambda| - (\mathcal{T})_{ii} \leq \sum_{j=1 \wedge j \neq i}^{n} |(\mathcal{T})_{ij}|$$

$$\iff 1 < |\lambda| \leq (\mathcal{T})_{ii} + \sum_{j=1 \wedge j \neq i}^{n} (\mathcal{T})_{ij} \ .$$

However, as $\mathcal{T}$ is strictly row-substochastic, we know that $(\mathcal{T})_{ii} + \sum_{j=1 \wedge j \neq i}^{n} (\mathcal{T})_{ij} < 1 < |\lambda|$ for all $i \in \underline{n}$. Thus, any strictly row-substochastic matrix $\mathcal{T}$ has no eigenvalues $|\lambda| > 1$. □

Definition 2.6 (Jordan normal form)
Every square matrix $A \in \mathbb{C}^{n\times n}$ is similar to a block diagonal matrix $J \in \mathbb{C}^{n\times n}$, also called the Jordan normal form of A, meaning $A = PJP^{-1}$ for an invertible matrix $P \in \mathbb{C}^{n\times n}$. The blocks on the diagonal of the Jordan matrix are called Jordan blocks J_i and have the form

$$J_i = \begin{bmatrix} \lambda_i & 1 & 0 & \dots & 0 \\ 0 & \lambda_i & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 & 0 \\ 0 & \ddots & 0 & \lambda_i & 1 \\ 0 & 0 & \dots & 0 & \lambda_i \end{bmatrix},$$

where λ_i is the i-th eigenvalue of A.

Theorem 2.7 (Convergence of strictly row-substochastic matrices)
If a matrix $\mathcal{T}$ is strictly row-substochastic, then it is convergent, i.e. $\lim_{k\to\infty} \mathcal{T}^k = 0$

Proof 2
Given a strictly row-substochastic matrix $\mathcal{T}$, consider its Jordan normal form $\mathcal{T} = PJP^{-1}$, see def. 2.6. Thus, $\lim_{k\to\infty} \mathcal{T}^k = P \lim_{k\to\infty} J^k P^{-1}$. Because $|\lambda| < 1$ for all eigenvalues of $\mathcal{T}$, $\lim_{k\to\infty} J^k = 0$. Thus, $\lim_{k\to\infty} \mathcal{T}^k = 0$, i.e. $\mathcal{T}$ is convergent.□

Definition 2.8 (Matrix inversion by the Neumann Series)
Given a matrix $\mathcal{T} \in \mathbb{R}^{n\times n}$ with $\lim_{k\to\infty} \mathcal{T}^k = 0$, i.e. $\mathcal{T}$ is convergent, the Neumann series $\sum_{k=0}^{\infty} \mathcal{T}^k$ converges against $(\boldsymbol{I} - \mathcal{T})^{-1}$.

Definition 2.9 (Irreducible and Reducible Matrix)
A non-negative square matrix $\mathcal{T} \in \mathbb{R}^{n\times n}_{\geq 0} > 0$ with $n \geq 2$ is irreducible if and only if $\sum_{k=0}^{n-1} \mathcal{T}^k > 0$. If $\mathcal{T}$ is not irreducible, it is reducible.

Definition 2.10 (Permutation Matrix)
A binary square matrix $P \in \{0,1\}^{n\times n}$ is a permutation matrix if and only if

(i) $(P)_{ij} = 1 = (P)_{i'j}$ with $i \neq i'$

(ii) $(P)_{ij} = 1 = (P)_{ij'}$ with $j \neq j'$

$(P^{-1} = P^{\mathsf{T}})$

In other words exactly one entry of P is 1 per row and column.

Definition 2.11 (Upper and Lower Triangular Matrix)
An upper/lower triangular matrix is a square matrix $\mathcal{T}_{up}, \mathcal{T}_{lo} \in \mathbb{R}^{n \times n}$, where all entries below/above the diagonal are 0.

$$\mathcal{T}_{up} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ & \ddots & \vdots \\ 0 & & a_{nn} \end{bmatrix} \quad \mathcal{T}_{lo} = \begin{bmatrix} a_{11} & & 0 \\ \vdots & \ddots & \\ a_{n1} & \dots & a_{nn} \end{bmatrix}$$

Definition 2.12 (LU-decomposition)
Let $\mathcal{T}$ be a square matrix, then (and only then) there exists an upper triangular matrix U and a lower triangular matrix L, such that

$$\mathcal{T} = L \cdot U \ .$$

By convention, L is made unitriangular, meaning the entries on its diagnonal are equal to 1. L & U invertible: all triangular matrices inv. (if all $a_{ii} \neq 0$)

Blocks $\mathcal{T}_{ii}$ communicating groups/classes (in graph) + $_{ij}$ communicating between groups i&j??

Definition 2.13
Invertible matrices have full rank

Expected Values

Definition 2.14 (Discrete Probability Distribution)
A function P is a *discrete probability distribution*, if $P : X \to [0, 1]$ where X is a countable set and $\sum_{x \in X} P(x) = 1$.

Definition 2.15 (Conditional Probability)
$Pr(A \mid B) = \frac{Pr(A \cap B)}{Pr(B)}$

Definition 2.16 (Law of Total Probability)
Let $B = \{B_1, B_2, \dots\}$ be a countable set of mutually exclusive and collectively exhaustive events, then for any event A,

$$Pr(A) = \sum_{i=1}^{|B|} Pr(A \mid B_i) \cdot Pr(B_i)$$

Definition 2.17 (Monotone Convergence Theorem)
...

Chapter 3

Markov Chains

In this section, we will introduce definitions and theorems for Markov Chains which we will later require for section on Interval Markov Chains. The notation is mostly based on [BK08], [CSH08], [BDLM21] and [MKQW25].

Definition 3.1 (DTMC)
A discrete-time Markov chain (DTMC) is a tuple $\mathcal{D} = (S, \mathbf{P}, \iota_{init})$, where

(i) S is a countable, nonempty set of states,

(ii) $\mathbf{P} : S \times S \to [0, 1]$ is the transition probability function, with $\sum_{s' \in S} \mathbf{P}(s, s') = 1$ for all states $s \in S$, and

(iii) $\iota_{init} : S \to [0, 1]$ is the initial distribution, such that $\sum_{s \in S} \iota_{init}(s) = 1$.

We assume in this thesis, that the DTMCs are finite, i.e., that the set of states S is finite. Furthermore, we assume an arbitrary but fixed linear order $\prec$ on the set S, such that the states $s \in S$ can be indexed. Thus, $s_i \prec s_j$ if $s, s' \in S$ and $i < j$. Due to the conditions (i) and (ii) provided in def. 3.1, the transition probability function $\mathbf{P}$ and the initial distribution ι_{init} are discrete probability distributions.

??(notion of a current state)

Moreover, if $\mathbf{P}(s, s') > 0$ for two given $s, s' \in S$, then it is possible to move from s to s' in one step. We refer to these states s' as direct successors of s (and to s as a direct predecessor of all states s'). Additionally, we define a neighbour set N, with $s' \in N(s)$, if and only if s' is a direct successor of s. For convenience, we define the set $S^{\mathcal{D}}_{in} = \{s \mid \iota^{\mathcal{D}}_{init}(s) > 0\}$.

Definition 3.2 (Transition Matrix)
The transition matrix $\mathcal{T}^{\mathcal{D}}$ of a DTMC $\mathcal{D}$ is a square matrix with size $|S| \times |S|$,

such that $(\mathcal{T}^{\mathcal{D}})_{ij} = \mathbf{P}(s_i, s_j)$.

Notice that due to the condition, given in (ii) of def. 3.1, that $\sum_{s_j \in S} \mathbf{P}(s_i, s_j) = 1$ for all states $s \in S$ and thus by defintion $\sum_{j=0}^{|S|} (\mathcal{T})_{ij} = 1$ for all rows $i \in \underline{n}$. In other words, the transition matrix $\mathcal{T}$ of a DTMC $\mathcal{D}$ is row-stochastic, see def. 2.3.

Theorem 3.3 (k-step Transition Probability)
The k-th power of the transition matrix $(\mathcal{T}^k)_{ij}$ characterizes the probability of transitioning from state i to j in precisely k steps, i.e., $Pr(\pi[k] = s_j \mid \pi[0] = s_i)$.

We prove this theorem after introducing further concepts/definitions in proof 4.

Definition 3.4 (Path of a DTMC)
An *infinite path* π in a DTMC $\mathcal{D}$ is a sequence $\pi = (s_0, s_1, \dots)$ of states with $s_0, s_1, \dots \in S$, such that $\iota_{init}(\pi[0]) > 0$ and $\mathbf{P}(\pi[k], \pi[k+1]) > 0$, where $\pi[k] \in S$ denotes the k-th element of the sequence/path for all $k \in \mathbb{N}_0$, i.e. $\pi[k] = s_k$. A *finite path* is a finite prefix of an *infinite path*.

Furthermore, we define the length $|\pi|$ of a finite path $\pi = (s_0, \dots, s_k) \in Paths^{\mathcal{D}}_{fin}$ to be equal to k and for an infinite path $\pi \in Paths^{\mathcal{D}}_{inf}$ to be $|\pi| = \infty$. The sets of all infinite and finite sets are denoted as $Paths^{\mathcal{D}}_{inf}$ and $Paths^{\mathcal{D}}_{fin}$, respectively.

Further, we construct the probability space of a DTMC as follows.

Definition 3.5 (Probability Space of a DTMC)
The *probability space* of a DTMC $\mathcal{D}$ is defined as $(Paths^{\mathcal{D}}_{inf}, \sum_{Paths^{\mathcal{D}}_{inf}}, Pr^{\mathcal{D}})$, where

- $\sum_{Paths^{\mathcal{D}}_{inf}}$ denotes the smallest σ-algebra containing the cylinder sets $Cyl(\pi')$ for all finite paths $\pi' \in Paths^{\mathcal{D}}_{fin}$ with

$$Cyl(\pi') = \{\pi \in Paths^{\mathcal{D}}_{inf} \mid \forall k \in \underline{|\pi'|} : \pi[k] = \pi'[k]\}$$

- $Pr^{\mathcal{D}}$ is the unique probability measure on $(Paths^{\mathcal{D}}_{inf}, \sum_{Paths^{\mathcal{D}}_{inf}})$ satisfying

$$Pr^{\mathcal{D}}(Cyl(\pi')) = \iota^{\mathcal{D}}_{init}(\pi'[0]) \cdot \prod_{k=0}^{|\pi'|-1} \mathbf{P}^{\mathcal{D}}(\pi'[k], \pi'[k+1])$$

  for all finite paths $\pi' \in Paths^{\mathcal{D}}_{fin}$

Additionally, we will use the notation $Pr_s(\pi)$ to denote the probability measure on the measurable space induced by fixing $s \in S$ as the unique initial state.

Definition 3.6 (Path Fragment)
Given a path π in a DTMC, a path fragment π' is an infix of π. In other words, a path fragment is a path with an arbitrary value for $\iota_{init}(\pi'[0])$.

Definition 3.7 (Intermediate state)
Given a DTMC $\mathcal{D}$, a set of states T, and a (goal) state $t \in S^{\mathcal{D}}$, then $u \in S^{\mathcal{D}}$ is intermediate if and only if there exists path fragment $\tilde{\pi}$ with $|\tilde{\pi}| = n$, see $\ldots$, such that $\tilde{\pi}[0] \in T, \tilde{\pi}[n] = s$ and $\tilde{\pi}[k] = u$ for some $k \in \underline{n}$. In other words there exists a path fragment from some $t \in T$ to s in $\mathcal{D}$ that contains u.

We denote the set of these intermediate states as $I^{\mathcal{D}}_{Ts}$ and I_{Ts} if the DTMC is clear from the context. Moreover, if T is a singleton set, i.e., $T = \{t\}$, then we simply denote the set of intermediate states as I_{ts}.

Definition 3.8 (Until Operator)
Let $R, T \subseteq S^{\mathcal{D}}$ be two sets of states of a DTMC $\mathcal{D}$. For these two sets, we define the "until" Operator as follows,

$$R \cup T = \{\pi \in Paths^{\mathcal{D}}_{inf} \mid \exists k \in \mathbb{N} : \pi[k] \in T \wedge \forall i < k : \pi[i] \in R\} \ .$$

In other words, the set $R \cup T$ contains all paths that only visit states in R, until a state in T is visited.

With the "until" operator, we define the reachability probability.

Definition 3.9 (Reachability Probability)
The reachability probability of a set $T \subseteq S^{\mathcal{D}}$ in a DTMC $\mathcal{D}$ is given by $Pr(\Diamond T)$, where $\Diamond T = S \cup T$. The operator $\Diamond$ is also called the "eventually" operator.

As $S^{\mathcal{D}}$ defines all states in $\mathcal{D}$, this simply defines the probability of visiting a state $t \in T$ at an arbitrary step in the future, i.e. if $Pr(\Diamond T) > 0$ at least one state $t \in T$ is reachable. In case $T = \{t\}$ is a singleton set, we omit the braces and denote $\Diamond t$ instead of $\Diamond\{t\}$ (and $S \cup t$ instead of $S \cup T$). Lastly, we denote $\Diamond_t T = \{\pi \in \Diamond T \mid \pi[0] = t\}$

$\sum_{Paths^{\mathcal{D}}_{inf}}$ DEFINIEREN (?)

Definition 3.10 (Characterizing Reachability Probability for Path Fragments)
For some set of paths $\Diamond S$ and an intial state t.

$$Pr^{\mathcal{D}}_t(\Diamond S) = \sum_{\pi \in \Diamond_t S} \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1])$$

Lemma 3.11 (Recursive Formula for Reachability Probability)
For states $t \in S$ and $T \subseteq S$:

$$Pr_t(\Diamond T) = \sum_{u \in N(t) \cap I_{tT}} \mathbf{P}(t,u) \cdot Pr_u(\Diamond T)$$

with $Pr_t(\Diamond T) = 1$ for all $t \in T$

Proof 3
According to def. 3.10,

$$\begin{aligned}
Pr_t^{\mathcal{D}}(\Diamond T) &= \sum_{\pi \in \Diamond_t T} \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) \\
&= \sum_{\pi \in \Diamond_t T} \mathbf{P}^{\mathcal{D}}(\pi[0], \pi[1]) \cdot \prod_{k=1}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) && (\pi[0] = t \text{ and } \pi[1] \in N(t)) \\
&= \sum_{u \in N(t)} \underbrace{\mathbf{P}(t,u)}_{=0, \text{ for all } u \notin I_{tT}^{\mathcal{D}}} \cdot \sum_{\pi \in \Diamond_u T} \prod_{k=1}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) \\
&= \sum_{u \in N(t) \cap I_{tT}^{\mathcal{D}}} \mathbf{P}(t,u) \cdot \sum_{\pi \in \Diamond_u T} \prod_{k=1}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) \\
&= \sum_{u \in N(t) \cap I_{tT}^{\mathcal{D}}} \mathbf{P}(t,u) \cdot Pr_u(\Diamond T)
\end{aligned}$$

□

Now that we have defined the probability space for DTMCs, we can prove theorem 3.3.

Proof 4
We will prove this theorem by induction.

Notice, that $Pr(\pi[k] = s_j \mid \pi[0] = s_i)$ for some $s_i, s_j \in S$ is precisely $Pr_{s_i}(\{\pi \in \Diamond_{s_i} s_j \mid |\pi| = k\})$. By defintion, see def. 3.2, $(\mathcal{T})_{ij} = \mathbf{P}(s_i, s_j) = Pr_{s_i}(\{\pi \in \Diamond_{s_i} s_j \mid |\pi| = 1\})$, thus the theorem holds for $k = 1$. Now, assume that the theorem holds for any $k \in \underline{n}$. As $(\mathcal{T}^{k+1})_{ij} = (\mathcal{T}^k \cdot \mathcal{T})_{ij} = \sum_{m=1}^{n} (\mathcal{T}^k)_{im} \cdot (\mathcal{T})_{mj}$. Using our induction hypothesis, $(\mathcal{T}^k)_{im}$ characterizes the probability of transition from state i to m in k steps, for all $m \in \underline{n}$

We use the LTL unary prefix operator ◯ to denote the temporal modality "next",

meaning

Definition 3.12 (Next Step Probability)
Let $\mathcal{D}$ be a DTMC and let $E \in \sum_{Paths_{inf}^{\mathcal{D}}}$ be a (countable) set of infinite paths in $\mathcal{D}$. We now define the "next step" operator, as

$$\bigcirc E = \{\pi \in Paths_{inf}^{\mathcal{D}} \mid (\pi[1], \pi[2], \dots) \in E\} \ ,$$

i.e., a path satisfies $\bigcirc E$ if and only if after removing the first state $\pi[0]$, the remaining path belongs to E. Moreover, the next step probability is defined as $Pr(\bigcirc E) = \sum_{s' \in N(s)} \mathbf{P}(s, s') \cdot Pr_{s'}(E)$.

In this thesis, we will use the "next step" operator $\bigcirc$ in combination with the "eventually" operator $\Diamond$. Specifically, we use them to denote the return probability $Pr_s(\bigcirc\Diamond s)$, i.e., the probability to reach s from all of its successors.

Definition 3.13 ((Bottom) Strongly Connected Component)
A set of states $C \subseteq S^{\mathcal{D}}$ of a DTMC $\mathcal{D}$ is strongly connected if and only if for every pair of states $s, t \in S$, there exists a path $\pi \in Paths_{inf}^{\mathcal{D}}$, such that $\pi[0] = s$ and $\pi[k]$ for some $k \in \mathbb{N}$. If C is strongly connected and no proper subset, i.e. $C' \subset C$, is strongly connected, then C is called a strongly connected component (SCC).

Furthermore, an SCC C is called a bottom stronlgy connected component (BSCC) if there exists no $k \in \mathbb{N}$ and no path $(\pi[0], \dots, \pi[k], \dots) \in Paths_{inf}^{\mathcal{D}}$ such that $\pi[0] \in C$ and $\pi[k] \notin C$.

In other words, once the model enters a BSCC, it cannot leave it again. Notice, that any grouping

SCC (minimal amount of states of set such that still strongly connec) (Acyclic DTMC)

The states can be classified ...

Definition 3.14 (Transient and Recurrent states)
A state $s \in S$ is called recurrent if and only if $Pr_s(\bigcirc\Diamond s) = 1$. All states that are not recurrent are called transient.

In other words, a state is recurrent precisely when we are guaranteed to reach it from all of its direct successors, meaning once we reach a recurrent state, we are bound to return to it. (Thus reaching it infinitely often) Using these definitions, we can split the states S of a DTMC into two sets with S_{tr} containing all of the transient states and S_{re} containing all of the recurrent states. Furthermore, we define $S_{tr \cap in} \subseteq S_{in}$ with $S_{tr \cap in} = S_{in} \cap S_{tr}$. Additionally, we define an absorbing state.

Definition 3.15 (Absorbing state)
A states s of a DTMC $\mathcal{D}$ is refered to as an absorbing state if and only if $\mathbf{P}^{\mathcal{D}}(s,s) = 1$.

This means, once the model enters an absorbing state, it cannot leave it. Notice, that absorbing states are a special type of recurrent states, as the only direct successor of an absorbing state $s \in S$ is s itself and therefore trivially $Pr_s(\Diamond s) = 1$.

Notice further, that intuitively the Markov Chain almost surely reaches a recurrent state in any DTMC.

Theorem 3.16 (Limit Behaviour)
For any DTMC $\mathcal{D}$, $s \in S$ with $\iota_{init}(s) > 0$ it holds that $Pr(\Diamond S_{re}) = 1$

Definition 3.17 (Acyclic DTMC)
A DTMC $\mathcal{D}$ is called acyclic if and only if each SCC is either a BSSC or a singleton set $\{s\}$ with $\mathbf{P}(s,s) = 0$

Definition 3.18 (Stopping Time)
A random variable $T : Paths^{\mathcal{D}}_{inf} \to \mathbb{N} \cup \{\infty\}$ is called a *stopping time* if, for every $n \geq \mathbb{N}$,

Definition 3.19 (Strong Markov Property)

Definition 3.20 (Markov Descision Process (MDP))
[BK08] A *Markov Decision Process* (MDP) is a tuple $\mathcal{M} = (S, Act, A, \mathbf{P}, \iota_{init})$, where

(i) S is a countable set of states

(ii) Act is a set of actions

(iii) $A(s) \subseteq Act$ denotes the applicable actions for all states $s \in S$

(iv) $\mathbf{P} : S \times Act \times S \to [0,1]$ is the transition probability function such that $\sum_{s' \in S} \mathbf{P}(s, \alpha, s') = 1$ for all $s \in S$ and $\alpha \in A(s)$

(v) $\iota_{init} : S \to [0,1]$ is the initial distribution such that $\sum_{s \in S} \iota_{init}(s) = 1$

Definition 3.21 (Canonical Form)
For a DTMC $\mathcal{D}$, its transition matrix $\mathcal{T}$ is said to be in canonical form if and only

if

$$\mathcal{T} = \begin{bmatrix} Q & R \\ 0 & S \end{bmatrix}$$

where

(i) $Q = (\mathbf{P}(s,t))_{s,t \in S_{tr}}$ contains the transition probabilities between the transient states

(ii) $R = (\mathbf{P}(s,t))_{s \in S_{tr}, t \in S_{re}}$ contains the transition probabilities between transient and recurrent state pairs

(iii) $S = (\mathbf{P}(s,t))_{s,t \in S_{re}}$ contains the transition probabilities between the recurrent states

For absorbing DTMCs $S = \boldsymbol{I}$. As mentioned in **??**, this is precisely possible for

(Reward Structure)

3.1 Expected Visiting Times for DTMCs

Intuitively, the notion of EVTs describes how often we expect the model to be in a given state. To formally introduce EVTs, we fix a DTMC $\mathcal{D}$ and start by introducing *visiting times* along paths.

Definition 3.22 (Visiting Times of Infinite Paths)
The function $vt_s : Paths^{\mathcal{D}}_{inf} \to \mathbb{R}$ describes the *visiting time* of a state $s \in S$ along infinite paths in $\mathcal{D}$, where for $\pi \in Paths^{\mathcal{D}}_{inf}$

$$vt_s(\pi) = \sum_{k=0}^{\infty} [\![\pi[k] = s]\!]$$

Borel Measurability

→ Thus, the Lebesgue integral of the non-negative function vt_s is well-defined for all states $s \in S^{\mathcal{D}}$. With this function, we can now define the EVT of a state s as the expected value of vt_s over all possible paths.

Definition 3.23 (Expected Visiting Times in DTMCs)
For a state $s \in S$, the *expected visiting time* of s is characterized as

$$\mathbb{E}^{\mathcal{D}}[vt_s] = \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_s(\pi) dPr^{\mathcal{D}}(\pi) \ .$$

Additionally, we denote the *expected visiting time* that is obtained by fixing a

certain state $t \in S$ as the only initial state as $\mathbb{E}_t^{\mathcal{D}}[vt_s]$. It is defined as

$$\mathbb{E}_t^{\mathcal{D}}[vt_s] = \int_{\pi \in Paths_{inf}^{\mathcal{D}} \wedge \pi[0]=t} vt_s(\pi) dPr^{\mathcal{D}}(\pi) \ .$$

As we defined the probability measure of paths π through a product containing the factor $\iota_{init}^{\mathcal{D}}(\pi[0])$ (see definition def. 3.5), it is nonzero only for paths that begin in a state s with $\iota_{init}^{\mathcal{D}}(s) > 0$. Thus the value $\mathbb{E}^{\mathcal{D}}[vt_s]$ gives the number of times the state s is expected to be visited from all paths in $\mathcal{D}$ that start in one of the initial states. Whenever it is clear from the context, we will omit the DTMC $\mathcal{D}$ from $\mathbb{E}^{\mathcal{D}}[vt_s]$ and $\mathbb{E}_t^{\mathcal{D}}[vt_s]$.

Using the aforementioned classification of the states of $\mathcal{D}$ into transient and recurrent states in def. 3.14, we can already give a characterization for the latter.

Lemma 3.24 (EVTs of Recurrent States)
For all recurrent states $s \in S_{re}$, it holds that

$$\mathbb{E}[vt_s] = \begin{cases} \infty & \text{if s is reachable,} \\ 0 & \text{else.} \end{cases}$$

For a proof, see [MKQW25].

Thus, we will focus on finding the non-trivial expected visiting times of the transient states.

We, will now see how the expected visiting times of the transient states can be calculated through a linear equation system. For this, we will first define the *fundamental matrix* which was introduced by Kemeny and Snell [KS76].

Definition 3.25 (Fundamental Matrix)
For a DTMC $\mathcal{D}$ with the matrix $Q = (\mathbf{P}^{\mathcal{D}}(s,t))_{s,t \in S_{tr}}$ containing the transition probabilities between the transient states (defined in def. 3.21), the fundamental matrix F is defined as,

$$F = (I - Q)^{-1} \ .$$

As Q is strictly row-substochastic, we can thus conclude using theorem 2.7, that Q is also convergent. Therefore, the Neumann series of Q converges with $\sum_{k=0}^{\infty} Q^k = (\boldsymbol{I} - Q)^{-1}$, i.e. the fundamental matrix (see def. 3.25) is well-defined.

Furthermore, in [KS76] Kemeny and Snell prove that each entry $(F)_{ij} = \mathbb{E}_i[vt_j]$. From this equation, it intuitively follows that transient states are only visited a finite amount of times. With this characterization of EVTs for transient states, we can prove the following lemma.

Lemma 3.26 (Reachability of Transient States)
Transient states are reachable only from other transient states. Moreover, it follows from this that if a state is reachable from a recurrent state, it must also be recurrent

We provide a short proof by contradiction

Proof 5
Assume that a transient state s_{tr} is reachable from a recurrent state s_{re}, i.e. $Pr_{s_{re}}(\bigcirc\Diamond s_{tr}) > 0$. As $Pr_{s_{re}}(\bigcirc\Diamond s_{re}) = 1$, see def. 3.14, necessarily s_{re} is reachable from s_{tr} as well and crucially $Pr_{s_{tr}}(\bigcirc\Diamond s_{re}) = 1$. We now view each visit of s_{re} as an indepenent trial, due to the strong Markov property def. 3.19. We can then describe the probability of not reaching s_{tr} as $\lim_{N\to\infty}(1 - Pr_{s_{re}}(\bigcirc\Diamond s_{tr}))^N = 0$, as s_{re} is visited infinitely often, see lemma 3.24. Thus, we are guarenteed to visit s_{tr}, when starting in s_{re}. As $Pr_{s_{tr}}(\bigcirc\Diamond s_{re}) = 1$, this process repeats each time s_{tr} is visited. Therefore, s_{tr} is visited infinitely often as well, i.e. $\mathbb{E}[vt_{s_{tr}}] = \infty$. This contradicts the characterization of the EVTs of transient states, proven by [KS76]. Thus, we conclude that transient states cannot be reachable from recurrent states, i.e. can only be reached from other transient states. □

Moreover, we can use this result to define a general characterization for the EVTs of transient states using the initial distribution.

Let us now define $\tau_{init} = (\iota_{init}(s_1), \dots, \iota_{init}(s_n))^\mathsf{T}$ for all $s \in S_{tr}$, so that we can rewrite **??** to $\mathbb{E}[vt_s] = (F^\mathsf{T})_s \cdot \tau_{init}$, where $(F^\mathsf{T})_s$ denotes the row vector of F^T corresponding to the state s. As introduced in def. 3.25, $(F)_s = ((I - Q)^{-1})_s$, from which we can conclude that $\mathbb{E}[vt_s] = ((I - Q^\mathsf{T})^{-1})_s \cdot \tau_{init}$ or $\mathcal{E} = (I - Q^\mathsf{T})^{-1} \cdot \tau_{init}$, where $\mathcal{E} = (\mathbb{E}[vt_{s_1}], \dots, \mathbb{E}[vt_{s_n}])^\mathsf{T}$. However, as the calculation of the inverse of a high-dimensional matrix becomes infeasible for large models, we will take a look at how the EVTs can be calculated without the fundamental matrix.

For this, we rewrite the equation $\mathcal{E} = (I - Q^\mathsf{T})^{-1} \cdot \tau_{init}$ to $\mathcal{E} = \tau_{init} + Q^\mathsf{T} \cdot \mathcal{E}$. For a single state $s \in S_{tr}$, we can thus calculate the respective EVT as follows,

$$\mathbb{E}[vt_s] = \iota_{init}(s) + \sum_{t \in S_{tr}} \mathbf{P}(t, s) \cdot \mathbb{E}[vt_t] \ . \tag{3.1}$$

3.1.1 Relating Expected Visiting Times and Quantitative Reachability

We will now relate *expected visiting times* to *quantitave reachability* for DTMCs.

Lemma 3.27
Given a DTMC $\mathcal{D}$ and its fundamental matrix F, we can define

$$(F)_{ts} = Pr_t(\Diamond s) \cdot (F)_{ss} \ ,$$

for all $s \neq t \in S_{tr}$, and thus

$$\mathbb{E}[vt_s] = Pr(\Diamond s) \cdot (F)_{ss} \ .$$

Trivially, all paths that reach s from t at least once visit s. Thus, intuitively, the *expected visiting time* of s starting in t is simply the product of the probability of reaching s and the *expected visiting time* of s starting in s.

Lemma 3.28
Given a DTMC $\mathcal{D}$ and its fundamental matrix F, we can define

$$(F)_{ss} = \frac{1}{1 - Pr_s(\bigcirc\Diamond s)}$$

for all $s \in S_{tr}$, where $Pr_s(\bigcirc\Diamond s) = \sum_{t \in N(s)} \mathbf{P}(s,t) \cdot Pr_t(\Diamond s)$

(The fraction is well-defined as $Pr_s(\bigcirc\Diamond s) < 1$, see def. 3.14)

Proof 6
As the only initial state is s, it is necessarily visited at least once. After that the probability that s will be visited at least one more time is $Pr_s(\bigcirc\Diamond s)$ and the probability that it will not, is accordingly $(1 - Pr_s(\bigcirc\Diamond s))$. Moreover, the probability of visiting s exactly k times is precisely the probability of returning $k-1$ times and not return again afterwards. Thus, $Pr_s(N_s = k) = Pr_s(\bigcirc\Diamond s)^{k-1}(1 - Pr_s(\bigcirc\Diamond s))$ for $k \in \mathbb{N}$, where N_s is a random variable representing the total number of time s is visited (Pr_s means starting in s). Intuitively,

$$\begin{aligned}
(F)_{ss} &= \sum_{k=1}^{\infty} k \cdot Pr_s(N_s = k) \\
&= \sum_{k=1}^{\infty} k \cdot Pr_s(\bigcirc\Diamond s)^{k-1}(1 - Pr_s(\bigcirc\Diamond s)) \\
&= (1 - Pr_s(\bigcirc\Diamond s)) \sum_{k=1}^{\infty} k \cdot Pr_s(\bigcirc\Diamond s)^{k-1} \\
&= (1 - Pr_s(\bigcirc\Diamond s)) \cdot \frac{1}{(1 - Pr_s(\bigcirc\Diamond s))^2} = \frac{1}{1 - Pr_s(\bigcirc\Diamond s)} \qquad \text{(by geometric series.)}
\end{aligned}$$

□

Finally, using **??**,

$$\begin{aligned}
\mathbb{E}[vt_s] &= Pr(\Diamond s) \cdot (F)_{ss} && \text{(by lemma 3.27)} \\
&= \frac{Pr(\Diamond s)}{1 - Pr_s(\bigcirc \Diamond s)} && \text{(by lemma 3.28)} \qquad (3.2) \\
&= \sum_{t \in S_{tr \cap in}} \frac{Pr_t(\Diamond s)}{1 - Pr_s(\bigcirc \Diamond s)} \cdot \iota_{init}(t)
\end{aligned}$$

Notice, that for any given DTMC $\mathcal{D}$ we can construct $\mathcal{D}^s_{uni} = (S^{\mathcal{D}} \cup \{s_{in}, s_{suc}\}, \mathbf{P}^{\mathcal{D}^s_{uni}}, \iota^{\mathcal{D}^s_{uni}}_{init})$, where $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_i, s_j) = \mathbf{P}^{\mathcal{D}}(s_i, s_j)$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s, s_j) = 0$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s, s_{suc}) = 1$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_{suc}, s_j) = \mathbf{P}^{\mathcal{D}}(s, s_j)$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_{in}, s_j) = \iota^{\mathcal{D}}_{init}(s_j)$ for all $s_i \in S^{\mathcal{D}} \backslash \{s\}$ and $s_j \in S^{\mathcal{D}}$, and $\iota^{\mathcal{D}^s_{uni}}_{init}(s_{in}) = 1$. Essentially, $\mathcal{D}^s_{uni}$ is a copy of $\mathcal{D}$, where instead of using ι_{init} as an initial distribution, we simply have a single initial state s_{in} and s has only one direct successor s_{suc}, which has the same probability distribution as s in $\mathcal{D}$.

Theorem 3.29
The reachability probabilities and thus also EVTs in $\mathcal{D}$ and the corresponding $\mathcal{D}^s_{uni}$ are identical.

Proof 7
First, we show that the reachability probabilities are equal (see def. 3.8) $Pr^{\mathcal{D}}(\Diamond T) = Pr^{\mathcal{D}^s_{uni}}(\Diamond T)$ for all $T \in S^{\mathcal{D}}$. For this, we define the bijection $\phi : Paths^{\mathcal{D}^s_{uni}}_{inf} \to Paths^{\mathcal{D}}_{inf}, \pi = (s_{in}, s_1, s_2, \dots, s_{k-1}, s_{suc}, s_k, \dots) \mapsto (s_1, s_2, \dots, s_{k-1}, s_k, \dots)$, i.e., the bijection ϕ removes s_{in} and all s_{suc}. This bijection is well defined, as $N(s_{in}) = \{s \mid \iota^{\mathcal{D}}_{init}(s) > 0\}$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_i, s_{suc}) > 0$ only for $s_i = s$, and $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_{suc}, s_i) = \mathbf{P}^{\mathcal{D}}(s, s_i)$ ($\to$ thus same (amount of) paths).

Moreover, assume a given $\pi \in Paths^{\mathcal{D}_{uni}}_{inf}$ has $\pi[k_i] = s_{suc}$ with $i \in \underline{j}$ for some

$j \in \mathbb{N}$ and $\phi(\pi) = \hat{\pi}$.

$$
\begin{aligned}
Pr^{\mathcal{D}_{uni}}(\pi) &= \underbrace{\iota_{init}^{\mathcal{D}_{uni}}(\pi[0])}_{=1} \cdot \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[k], \pi[k+1]) \\
&= \mathbf{P}^{\mathcal{D}_{uni}}(\pi[0], \pi[1]) \cdot \underbrace{\prod_{i=1}^{j-1} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[\underbrace{k_i - 1}_{=s}], \pi[\underbrace{k_i}_{=s_{suc}}])}_{=1} \cdot \prod_{k=1}^{k_1-2} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[k], \pi[k+1]) \\
&\quad \cdot \prod_{k=k_1}^{k_2-2} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[k], \pi[k+1]) \cdot \ldots \cdot \prod_{k_j}^{|\pi|-1} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[k = k_j], \pi[k+1]) \\
&= \iota_{init}^{\mathcal{D}}(\phi(\pi)[1]) \cdot \prod_{k=1}^{|\hat{\pi}|-1} \mathbf{P}^{\mathcal{D}}(\phi(\pi)[k], \phi(\pi)[k+1]) \qquad (\text{as, } \pi[k_i + 1] = \hat{\pi}[k_i]) \\
&= \iota_{init}^{\mathcal{D}}(\hat{\pi}[0]) \cdot \prod_{k=0}^{|\hat{\pi}|-1} \mathbf{P}^{\mathcal{D}}(\hat{\pi}[k], \hat{\pi}[k+1]) \\
&= Pr^{\mathcal{D}}(\hat{\pi}) \ .
\end{aligned}
$$

Therefore, we can conclude $Pr^{\mathcal{D}_{uni}}(\Diamond T) = Pr^{\mathcal{D}}(\Diamond T)$, as $\Diamond T \subseteq Paths_{inf}^{\mathcal{D}_{uni}}$. Using eq. (3.2), we can then also conclude that the EVTs of $\mathcal{D}$ and $\mathcal{D}_{uni}$ must be equal.□

Thus, the reachability probabilities $Pr_{s_{in}}^{\mathcal{D}_{uni}}(\Diamond s) = \sum_{t \in S_{in}} Pr_t^{\mathcal{D}}(\Diamond s) = \sum_{t \in S_{tr} \cap in} Pr_t^{\mathcal{D}}(\Diamond s)$ are equal and we can simplify the previous expression,

$$
\mathbb{E}^{\mathcal{D}}[vt_s] = \frac{Pr_{s_{in}}^{\mathcal{D}_{uni}}(\Diamond s)}{1 - Pr_{s_{suc}}^{\mathcal{D}_{uni}}(\Diamond s)} \cdot \tau_{init}(s_{in}) \left(= \frac{Pr^{\mathcal{D}}(\Diamond s)}{1 - Pr_s^{\mathcal{D}}(\bigcirc \Diamond s)} \cdot \tau_{init}(s_{in}) \right) \ . \tag{3.3}
$$

In other words, we can calculate the EVT of a DTMC $\mathcal{D}$ with only two reachability probabilities using the corresponding DTMC $\mathcal{D}_{uni}$.

Moreover, using lemma 3.27 and the construction $\mathcal{D}_{uni}$, we can calculate the quantitative reachability of a given state with $Pr(\Diamond s) = \frac{(F^{\mathcal{D}_{uni}})_{s_{in}s}}{(F)_{ss}}$.

Chapter 4

Interval Markov Chains

The focus of this thesis are expected visiting times in Interval Markov Chains, which we will introduce in this chapter. In the following sections we will mainly discuss how the expected visiting times can be calculated and shortly examine stationary distributions of Interval Markov Chains.

We start by introducing concepts which we will use in the following sections.

Definition 4.1 (Interval Markov Chain)
An interval discrete-time Markov chain (IDTMC) is a tuple $\mathcal{I} = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$, where

(i) S is a countable, nonempty set of states,

(ii) $\underline{\mathbf{P}} : S \times S \to [0,1]$ is the lower transition probability function, where $\underline{\mathbf{P}}(s, s')$ gives the lower bound on the transition probability from state s to s',

(iii) $\overline{\mathbf{P}} : S \times S \to [0,1]$ is the upper transition probability function, where $\overline{\mathbf{P}}(s, s')$ gives the upper bound on the transition probability from state s to s',

(iv) $\underline{\iota}_{init} : S \to [0,1]$ is the lower bound on the initial distribution, and

(v) $\overline{\iota}_{init} : S \to [0,1]$ is the upper bound on the initial distribution.

Definition 4.2 (Path of an IDTMC)
Similarily to DTMCs, an *infinite path* in an IDTMC $\mathcal{I}$ is a sequence $\pi = (s_0, s_1, \dots)$ of states with $s_0, s_1, \dots \in S^{\mathcal{I}}$, such that $\underline{\mathbf{P}}(\pi[k], \pi[k+1]) > 0$ for all $k \in \mathbb{N}$.

To resolve the uncertainty caused by the intervals, we need to define semantics to interpret the Interval Markov Chain. In other words we need to resolve an Interval Markov Chain to some set of Markov Chains that correspond to it. We say that all of these DTMCs $\mathcal{D}_i$ satisfy the IDTMC $\mathcal{I}$ and denote it as $\mathcal{D}_i \models \mathcal{I}$. We assume in this

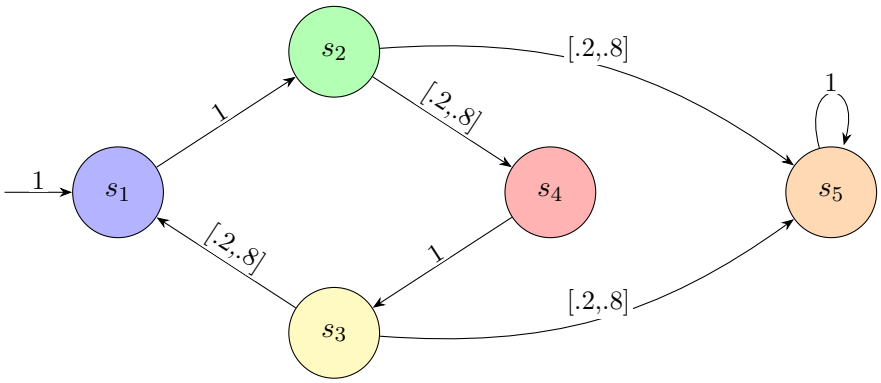


Figure 4.1: IDTMC $\mathcal{I}$

thesis that there exists at least one satisfying DTMC for every IDTMC.

Assumption 1
In sec. 3.1, we concluded that due to lemma 3.24, only the transient states need to be considered for calculating EVTs. As we are interested in the calculation of EVTs, we will only IDTMCs with a finite amount of transient states. Moreover, we will only consider IDTMCs where all states are reachable in at least one satisfying IDTMC, as if they are not the EVTs of those states are trivially 0 for all satisfying DTMCs and have no impact on the EVTs of other states.

In this paper, we will consider two popular semantics: the *once-and-for-all*, also know as the *Uncertain Markov Chain (UMC) semantic*, and the *IMDP semantic*. We will first define the former, similar to [BDLM21, CSH08, BLW13].

Definition 4.3 (Once-and-for-all/UMC Semantic)
An arbitrary DTMC $\mathcal{D}$ satisfies an IDTMC $\mathcal{I}$ if and only if $\underline{\mathbf{P}}^{\mathcal{I}}(s,s') \leq \mathbf{P}^{\mathcal{D}}(s,s') \leq \overline{\mathbf{P}}^{\mathcal{I}}(s,s')$ and $\underline{\iota}^{\mathcal{I}}_{init}(s) \leq \iota^{\mathcal{D}}_{init}(s) \leq \overline{\iota}^{\mathcal{I}}_{init}(s)$ for all $s, s' \in S^{\mathcal{I}} = S^{\mathcal{D}}$.

We will denote this as $\mathcal{D} \models_{\mathrm{U}} \mathcal{I}$.

In other words, any DTMC whose initial distribution and transition probabilities lie in the intervals defined by the IDTMC is satisfactory. Moreover, notice that the satisfaying DTMCs have the same state size (amount of states) as the IDTMC.

Example 4.4
Consider the IDTMC $\mathcal{I}$ in fig. 4.1. The DTMC $\mathcal{D}_U$ depicted in fig. 4.2 satisfies $\mathcal{I}$ under the UMC semantic, i.e., $\mathcal{D}_U \models_U \mathcal{I}$.

The second semantic which we will define, is the *IMDP semantic*. There are two

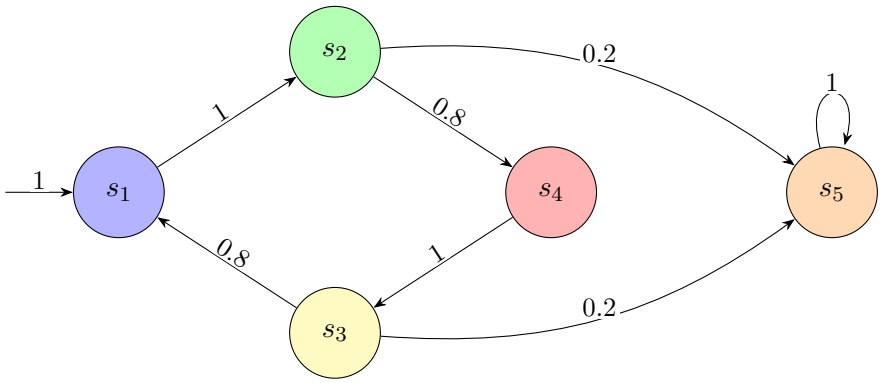


Figure 4.2: DTMC $\mathcal{D}_U$ with $\mathcal{D}_U \models_U \mathcal{I}$

common definitions for this semantic. In this thesis, we will use the definition found in [SVA06] and [CSH08].

Definition 4.5 (IMDP semantic)
We define $Steps(s)$ as the set of probability functions over $S^{\mathcal{I}}$, i.e. $Steps(s) = \{\mu : S \to [0,1] \mid \sum_{t \in S} \mu(t) = 1 \text{ and } \underline{\mathbf{P}}(s,t) \leq \mu(t) \leq \overline{\mathbf{P}}(s,t) \text{ for all } t \in S\}$ for all $s \in S^{\mathcal{I}}$. Let $last(\pi)$ denote the last state of a finite path $\pi \in Paths^{\mathcal{I}}_{fin}$, i.e., if $|\pi| = n$, then $last(\pi) = \pi[n]$.

The choice of a transition function is made by *nature*. A *nature* for an IDTMC $\mathcal{I}$ is a function $\eta : Paths^{\mathcal{I}}_{fin} \to Steps$ that assigns each finite path to a possible probability transition function, i.e., $\eta(\pi) \in Steps(last(\pi))$.

In other words, for any given finite path $\pi \in Paths^{\mathcal{I}}_{fin}$, the *nature* chooses a probability distribution from $Steps(last(\pi))$ according to which the successor state is then sampled. (Still need to define $Paths_{fin}$ for IDTMCs)

The other common definition can be found in [BDL+17] and [BDLM21].

Example 4.6
Consider the IDTMC $\mathcal{I}$ in fig. 4.1. The DTMC $\mathcal{D}_I$ depicted in fig. 4.3 satisfies $\mathcal{I}$ under the IMDP semantic, i.e., $\mathcal{D}_I \models_I \mathcal{I}$.

As the IMDP semantic allows for a probability distribution to be chosen in each state at every step, we can interpret an IDTMC under the IMDP semantic as a Markov Decision Process (MDP), see def. 3.20, as the notion of *nature* is identical to the notion of *schedulers* used to resolve the uncertainty in MDPs.

As defined in [BK08].

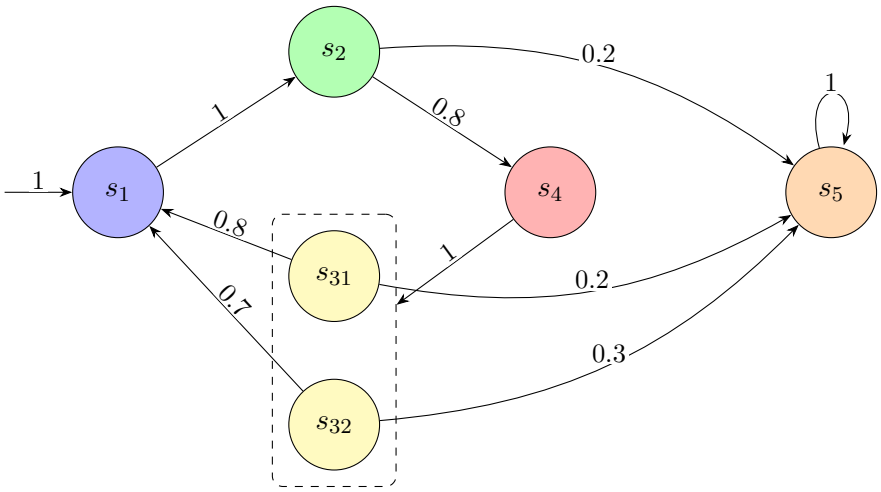


Figure 4.3: DTMC $\mathcal{D}_I$ with $\mathcal{D}_I \models_I \mathcal{I}$.

Definition 4.7 (Scheduler)
Let $\mathcal{M} = (S, Act, A, \mathbf{P}, \iota_{init})$ be a MDP. A *scheduler* for $\mathcal{M}$ is a function $\mathfrak{S} : Paths^{\mathcal{M}}_{fin} \to Act$ such that $\mathfrak{S}(s_0, \ldots, s_n) \in A(s_n)$.

By resolving the uncertainty of an MDP, a scheduler induces a Markov Chain. In [BK08] (Definition 10.92) these induced Markov Chains are defined as follows

Definition 4.8 (Markov Chain of an MDP Induced by a Scheduler)
For an MDP $\mathcal{M} = (S, Act, A, \mathbf{P}, \iota_{init})$ and a scheduler $\mathfrak{S}$ on $\mathcal{M}$, the induced Markov Chain $\mathcal{D}_{\mathfrak{S}}$ is given by

$$\mathcal{M}_{\mathfrak{S}} = (Paths^{\mathcal{M}}_{inf}, \mathbf{P}_{\mathfrak{S}}, \iota_{init})$$

where for $\sigma = s_0, s_1, \ldots, s_n$:

$$\mathbf{P}_{\mathfrak{S}}(\sigma, \sigma s_n) = \mathbf{P}(s_n, \mathfrak{S}(\sigma), s_{n+1}) \ .$$

These induced Markov Chains are precisely the satisfying Markov Chains. Notice, that even if the MDP is finite, any induced Markov Chain is infinite.

The most intuitive reduction from an IDTMC $\mathcal{I}$ under the IMDP semantic to an MDP $\mathcal{M}$ is to define $S^{\mathcal{M}} = S^{\mathcal{I}}$ and an uncoutable set $\bigcup_{s \in S} A(s) = Act$ of actions where each action $\alpha \in A(s)$ corresponds to a probability distribution in $Steps(s)$. In other words, the actions $\alpha \in A(s)$ in a given state $s \in S$ correspond to the probability distributions that can be chosen in s in any given step. However, as these subsets $A(s)$ can be uncoutable, this reduction is not useful (as no algorithm like value or policy iteration would terminate).

Instead, it is thus argued in [SVA06], [CSH08], and [CHK13], that the local constraints of each state of these MDPs and IDTMCs, see in def. 3.20 $(iv) - (v)$ and def. 4.1 $(ii) - (v)$, can be used to limit the amount of probability distributions that need to be mapped. In the theory of linear programming, this is called a basic feasible solution (BFS). For this reduction, the sets $A(s)$ with $s \in S^{\mathcal{I}}$ are exponential in the size of $|S^{\mathcal{I}}|$.

(Gleiche reach. prob. für diese Konstrukion in paper gezeigt (denke ich), weil reach prob $\subset$ PCTL. Und die Konstruktion ist für PCTL?)

Important for this thesis are in particular memoryless schedulers. The following definition is taken from [BK08].

Definition 4.9 (Memoryless Scheduler)
Let $\mathcal{M}$ be a MDP with state space S. Any scheduler $\mathfrak{S}$ on $\mathcal{M}$ is *memoryless* if and only if for each sequence $(s_0, s_1, \ldots, s_n)$ and $(t_0, t_1, \ldots, t_m)$ with $s_n = t_m$

$$\mathcal{D}(s_0, s_1, \ldots, s_n) = \mathcal{D}(t_0, t_1, \ldots, t_m) \ .$$

Thus, in this case, $\mathfrak{S}$ can be viewed as a function $\mathfrak{S} : S \to Act$.

In other words, if a scheduler is memoryless, then it always selects the same action in a given state. Notice, that in regards to the MDP construction in [SVA06] and [CSH08], where the actions represent the chosen probability distribution, this describes precisely the UMC semantic. As a memoryless scheduler chooses the same action in a given state, it chooses the same probability distribution for a given state in the MDP construction.

Definition 4.10 (Markov Chain of an MDP Induced by a memoryless Scheduler)
For an MDP $\mathcal{M} = (S, Act, A, \mathbf{P}, \iota_{init})$ and a scheduler $\mathfrak{S}$ on $\mathcal{M}$, the induced Markov Chain $\mathcal{D}_{\mathfrak{S}}$ is given by

$$\mathcal{M}_{\mathfrak{S}} = (S, \mathbf{P}_{\mathfrak{S}}, \iota_{init})$$

where for any $s_i, s_j \in S$:

$$\mathbf{P}_{\mathfrak{S}}(s_i, s_j) = \mathbf{P}(s_i, \mathfrak{S}(s_i), s_j) \ .$$

As $\underline{\mathbf{P}}(s_i, s_j) \leq \mathbf{P}(s_i, \mathfrak{S}(s_i), s_j) \leq \overline{\mathbf{P}}(s_i, s_j)$ by construction, the induced Markov Chain of a memoryless scheduler on the MDP construction satisfies all conditions of def. 4.3.

Often in the literature, see for example [CHK13] (continue list), the authors do not define a seperate probability function for the initial distribution, as we have done, but instead only define a singular initial state s_0. Notice however, that our definition can easily be fitted into this definition and vice versa, by mapping our initial distribution onto the transition probability function of the state s_0, similarly as in sec. 3.1.1 with

$\mathcal{D}_{uni}$ and s_{in}. We will refer to this definition as the fixed-initial-state-definition.

(proof 7 shown that reach prob identical –¿ and thus, as shown later, EVT; relevant??)

If we want to emphasize that a DTMC $\mathcal{D}$ satisfies the IDTMC $\mathcal{I}$ under both semantics, we write $\mathcal{D} \models \mathcal{I}$. Furthermore, if two DTMCs $\mathcal{D}_1$ and $\mathcal{D}_2$ satisfy a given IDTMC $\mathcal{I}$, we write $\mathcal{D}_1, \mathcal{D}_2 \models \mathcal{I}$ for short.

We use the definition of the quantitative reachability property in IDTMCs provided in [BDL+17].

Definition 4.11 (Reachability Probability in IDTMCs)
Given an IDTMC $\mathcal{I}$ and a set of states $T \subseteq S$, then a quantitative reachability property on $\mathcal{I}$ is defined as $Pr^{\mathcal{I}}(\Diamond T) \sim p$, where $0 \leq p \leq 1$ and $\sim \in \{\leq, <, >, \geq\}$. This quantitative property is satisfied if and only if there exists a DTMC $\mathcal{D} \models \mathcal{I}$ such that $Pr^{\mathcal{D}}(\Diamond T) \sim p$.

Definition 4.12 (Transient and Recurrent states in IDTMCs)
Similar to DTMCs, a state $s \in S^{\mathcal{I}}$ is called recurrent if and only if $Pr_s^{\mathcal{I}}(\bigcirc\Diamond s) = 1$, i.e., there exists a satisfying DTMC $\mathcal{D} \models \mathcal{I}$ such that $Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) = 1$. All states that are not recurrent are called transient.

Definition 4.13 (BSCCs for IDTMCs)
Let $\mathcal{I}$ be an IDTMC, then $B^{\mathcal{D}} \subseteq S^{\mathcal{I}}$ is a BSSC, if $B^{\mathcal{D}}$ is a BSCC in the DTMC $\mathcal{D}$, see def. 3.13, and $\mathcal{D} \models \mathcal{I}$.

As defined in [BK08].

Definition 4.14 (History-dependent Scheduler)

4.0.1 Relation between the semantics

Clearly, the IMDP semantic is more expressive than the UMC semantic [BDLM21]. More specifically for any given IDTMC $\mathcal{I}$, a DTMC $\mathcal{D}$ that satisfies $\mathcal{I}$ according to the UMC semantic, i.e. $\mathcal{D} \models_U \mathcal{I}$, does also satisfy $\mathcal{I}$ according to the IMDP semantic, i.e. $\mathcal{D} \models_I \mathcal{I}$ [BDLM21].

Theorem 4.15
For any IDTMC $\mathcal{I}$, if $\mathcal{D} \models_U \mathcal{I}$, then $\mathcal{D} \models_I \mathcal{I}$

For this, we provide a slightly more detailed proof of [BDLM21] (Remark 1).

Proof 8
If $\mathcal{D} \models_U \mathcal{I}$, then we define $T = S$ and $\tau(s) = s$. Then (i) of def. 4.5 is fulfilled as τ is bijective and $\iota_{init}(s) \in [\underline{\iota}_{init}(s), \overline{\iota}_{init}(s)]$ for all $s \in S$, as $\mathcal{D} \models_U \mathcal{I}$. Moreover, (ii) is fulfilled as $\mathbf{P}(s,t) \in [\underline{\mathbf{P}}(s,t), \overline{\mathbf{P}}(s,t)]$ for all $s, t \in S$, as $\mathcal{D} \models_U \mathcal{I}$. Thus, we can conclude that $\mathcal{D} \models_I \mathcal{I}$. □

Importantly, the two semantics have the same maximum and minimum reachability probability for any given (initial) state, according to [BDL+17] and [CHK13].

4.1 Expected Visiting Times in IDTMCs

We will now define *expected visiting times* (EVTs) for IDTMCs. As we discussed in the previous section, IDTMCs can be seen as a set of DTMCs that satiesfy the IDTMC according to the given semantic. These different DTMCs can have varying EVTs for a given state s in the IDTMC, thus we are interested in the upper and lower bound for the EVTs of s, i.e., in the highest and lowest EVT of s among the satisfying DTMCs.

Definition 4.16 (EVTs of IDTMCs)
For a IDTMC $\mathcal{I}$, we define the *upper expected visiting time* (uEVT) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ and *lower expected visiting time* (ℓEVT) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ for all states $s \in S^{\mathcal{I}}$, as follows for the UMC semantic

(i) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \inf_{\mathcal{D} \models_U \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_s]$

(ii) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \sup_{\mathcal{D} \models_U \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_s]$

and the IMDP semantic

(i) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \inf_{\mathcal{D} \models_I \mathcal{I}} \sum_{t \in T \wedge \tau^{\mathcal{D}}(t) = s} \mathbb{E}^{\mathcal{D}}[vt_t]$

(ii) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \sup_{\mathcal{D} \models_I \mathcal{I}} \sum_{t \in T \wedge \tau^{\mathcal{D}}(t) = s} \mathbb{E}^{\mathcal{D}}[vt_t]$.

4.1.1 Relating Expected Visiting Times and Quantitative Reachability

Similar to sec. 3.1.1, we will now relate *expected visiting times* to *quantitave reachability*. Moreover, we show that for any IDTMC $\mathcal{I}$, there exists a DTMC $\mathcal{D} \models \mathcal{I}$ that has the optimal, i.e., maximum or minimum, reachability probability for all transient states in $\mathcal{I}$.

Given the quantitative reachabilities of the DTMCs that satisfy a certain IDTMC $\mathcal{I}$, we can calculate the corresponding EVTs of a given state s in $\mathcal{I}$ and thus find the

lEVTs and uEVTs using the results from sec. 3.1.1.

As we concluded in sec. 3.1.1, we can calculate the EVT of a given state s in a DTMC $\mathcal{D}$ via quantitative reachability, using the corresponding DTMC $\mathcal{D}_{uni}$ and the eq. (3.2) $\mathbb{E}^{\mathcal{D}}[vt_s] = \sum_{t \in S_{tr} \cap in} \frac{Pr_t^{\mathcal{D}}(\Diamond s)}{1 - Pr_s^{\mathcal{D}}(\bigcirc \Diamond s)} \cdot \iota_{init}(t)$. Thus, the uEVT and ℓEVT of a given IDTMC $\mathcal{I}$ are given by the largest and smallest ratio of these quantitative reachability probabilites among the satisfying Markov Chains.

Remark. *Existence of DTMC with optimal reachability.* For every transient state $t \in S_{tr}$ there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ with $Pr_t^{\mathcal{D}}(\Diamond s) = \sup_{\mathcal{D}_i \models_U \mathcal{I}} Pr_t^{\mathcal{D}_i}(\Diamond s)$. This follows directly from the compactness and continuity of the polytope of DTMCs satisfying $\mathcal{I}$; it is not reproved here, only reused. To shorten the notation we will denote $\overline{Pr_t}^{\mathcal{D}}(\Diamond s) = \sup_{\mathcal{D}_i \models_U \mathcal{I}} Pr_t^{\mathcal{D}_i}(\Diamond s)$ and $\underline{Pr_t}^{\mathcal{D}}(\Diamond s) = \inf_{\mathcal{D}_i \models_U \mathcal{I}} Pr_t^{\mathcal{D}_i}(\Diamond s)$.

Theorem 4.17
Given an IDTMC $\mathcal{I}$, a transient state $t \in S^{\mathcal{I}}$, a (goal) state $s \in S^{\mathcal{I}}$, and a DTMCs $\mathcal{D}$ with $\mathcal{D} \models \mathcal{I}$. (Positive transition/reachability probability from t to s, i.e. there needs to exist a path fragment from t to s)

If $\mathcal{D}$ has the maximum reachability probability from t to s, i.e., $Pr_t^{\mathcal{D}}(\Diamond s) = \overline{Pr_t}^{\mathcal{I}}(\Diamond s)$, then for all $u \in I_{ts}^{\mathcal{D}}$:

$$Pr_u^{\mathcal{D}}(\Diamond s) = \overline{Pr_u}^{\mathcal{I}}(\Diamond s)$$

Proof 9
We will start by proving this theorem for $\mathcal{D} \models_I \mathcal{I}$ and then show why $\mathcal{D} \models_U \mathcal{I}$ follows from it.

As shown in [BK08] (Theorem 10.100, pg. 851), for any MDP $\mathcal{M}$ the maximum reachability probability is characterized by the equation

$$\overline{Pr_t}^{\mathcal{I}}(\Diamond s) = \max \left\{ \sum_{u \in N(t)} \mathbf{P}(t, \alpha, u) \cdot \overline{Pr_u}^{\mathcal{I}}(\Diamond s) \mid \alpha \in Act(t) \right\} . \tag{4.1}$$

□

Clearly, the policies which solve this equation have the maximum reachability probability $Pr_u(\Diamond s)$ for all $u \in I_{ts}^{\mathcal{D}}$, as shown in [BK08]. As shown in the previous section, an IDTMC under the IMDP semantic can be mapped to a MDP, where the policy represents a given satisfying DTMC (bzw. nach einer/der üblichen IMDP Semantiken ist eine policy/adversary das satisfying DTMC). Thus, this theorem is proven for the IMDP semantic. (Also importantly, the solution to the set of equations is unique, as shown in [BK08]. Thus, the maximum reachability probabilities of the states are unique)

As proven in [BK08] (Lemma 10.102 and Lemma 10.113), given any finite MDP $\mathcal{M}$ with state space S, and $T \subseteq S$, there exists *memoryless* schedulers $\mathfrak{S}_{max}$ and $\mathfrak{S}_{min}$ for every $t \in S$, such that $\overline{Pr}_t^{\mathcal{M}}(\Diamond T) = Pr_t^{\mathfrak{S}_{max}}(\Diamond T)$ and $\underline{Pr}_t^{\mathcal{M}}(\Diamond T) = Pr_t^{\mathfrak{S}_{min}}(\Diamond T)$. In other words, for every MDP, there exist *memoryless* schedulers that has maximum or minimum reachability to T from t. As discussed, a memoryless scheduler on the MDP construction induces a DTMC that satisfies the IDTMC. Because this memoryless scheduler is optimal, clearly it satisfies eq. (4.1). Thus, this theorem is proven for the UMC semantic.

Corollary 4.18
Given an IDTMC $\mathcal{I}$, two transient states $t_1, t_2 \in S^{\mathcal{I}}$, a (goal) state $s \in S^{\mathcal{I}}$, and two DTMCs $\mathcal{D}_1, \mathcal{D}_2$ with $\mathcal{D}_1, \mathcal{D}_2 \models_U \mathcal{I}$ and $Pr_{t_1}^{\mathcal{D}_1}(\Diamond s) = \overline{Pr}_{t_1}^{\mathcal{I}}(\Diamond s)$ and $Pr_{t_2}^{\mathcal{D}_2}(\Diamond s) = \overline{Pr}_{t_2}^{\mathcal{I}}(\Diamond s)$. In other words the DTMCs have a maximum reachability probability for t_1 and t_2, respectively.

For all $u \in I_{t_1 s}^{\mathcal{D}_1} \cap I_{t_2 s}^{\mathcal{D}_2}$, the reachability probabilities $Pr_u^{\mathcal{D}_1}(\Diamond s) = Pr_u^{\mathcal{D}_2}(\Diamond s)$ are identical.

Proof 10
This corollary follows from theorem 4.17. Using theorem 4.17, we can conclude that $Pr_u^{\mathcal{D}_1}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t_1 s}^{\mathcal{D}_1}$ and $Pr_u^{\mathcal{D}_2}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t_2 s}^{\mathcal{D}_2}$. Thus, $Pr_u^{\mathcal{D}_1}(\Diamond s) = Pr_u^{\mathcal{D}_2}(\Diamond s)$ for all $u \in I_{t_1 s}^{\mathcal{D}_1} \cap I_{t_2 s}^{\mathcal{D}_2}$. □

Lemma 4.19 (One-state Extension)
Given a set $T \subset S$, a state $t \notin T$, and a (goal) state s, assume that $\mathcal{D}_T \models_U \mathcal{I}$ with $Pr_u^{\mathcal{D}_T}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T$ and $\mathcal{D}_t \models_U \mathcal{I}$ with $Pr_t^{\mathcal{D}_t}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$. Define $\mathcal{D}_{cmb} \models_U \mathcal{I}$ by

$$\mathbf{P}^{\mathcal{D}_{cmb}}(w, v) = \begin{cases} \mathbf{P}^{\mathcal{D}_t}(w, v) & \text{if } w \in I_{ts}^{\mathcal{D}_t} \backslash I_{Ts}^{\mathcal{D}_T} \\ \mathbf{P}^{\mathcal{D}_T}(w, v) & \text{else.} \end{cases}$$

Stated in words, $\mathcal{D}_{cmb}$ is constructed such that its probability distribution function is equal to that of $\mathcal{D}_t$ in all states that are only intermediate states for t and not T, while it is equal to that of $\mathcal{D}_T$ in all other states.

Then, $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_u^{\mathcal{D}_{cmb}}(\Diamond s)$ for all $u \in T \cup \{t\}$, i.e., $\mathcal{D}_{cmb}$ has the maximum reachability probability for all states in $T \cup \{t\}$.

Proof 11
As $\mathbf{P}^{\mathcal{D}_{cmb}}(w, v) = \mathbf{P}^{\mathcal{D}_T}(w, v)$ for all $w \in I_{Ts}^{\mathcal{D}_T}$, clearly $\overline{Pr}_u^{\mathcal{I}}(\Diamond s) = Pr_u^{\mathcal{D}_{cmb}}(\Diamond s)$ for

all $u \in T$, see lemma 3.11.

Thus, we need to show that $Pr_t^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$. As we know that $Pr_t^{\mathcal{D}_t}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$, we will prove that $Pr_t^{\mathcal{D}_{cmb}}(\Diamond s) = Pr_t^{\mathcal{D}_t}(\Diamond s)$. Notice, that if $I_{ts}^{\mathcal{D}_t} \cap I_{Ts}^{\mathcal{D}_T} = \emptyset$, then the proof concludes using the same argument as for $u \in T$. Furthermore, if $I_{ts}^{\mathcal{D}_t} \subseteq I_{Ts}^{\mathcal{D}_T}$, then the proof concludes directly using theorem 4.17.

Recall, that $Pr_t^{\mathcal{D}_{cmb}}(\Diamond s) = \sum_{u \in N(t) \cap I_{ts}^{\mathcal{D}_{cmb}}} \mathbf{P}^{\mathcal{D}_{cmb}}(t,u) \cdot Pr_u^{\mathcal{D}_{cmb}}(\Diamond s)$, see lemma 3.11. For all $u \in I_{ts}^{\mathcal{D}_t} \setminus I_{Ts}^{\mathcal{D}_T}$, $\mathbf{P}^{\mathcal{D}_{cmb}}(u,w) = \mathbf{P}^{\mathcal{D}_t}(u,w)$ and for all $u \in I_{Ts}^{\mathcal{D}_T} \cap I_{ts}^{\mathcal{D}_t}$, $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = Pr_u^{\mathcal{D}_t}(\Diamond s)$, see corollary 4.18. From this, we can conclude that $Pr_t^{\mathcal{D}_t}(\Diamond s) = Pr_t^{\mathcal{D}_{cmb}}(\Diamond s)$. Thus, we have shown that $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_u^{\mathcal{D}_{cmb}}(\Diamond s)$ for all $u \in T \cup \{t\}$.

Into only remains to show, that $\mathcal{D}_{cmb} \models_U \mathcal{I}$. As $\mathcal{D}_t, \mathcal{D}_T \models_U \mathcal{I}$ it clearly follows from the construction that also $\mathcal{D}_{cmb} \models \mathcal{I}$. □

Crucially, lemma 4.19 can be extended to all transient states.

Theorem 4.20
Given an IDTMC $\mathcal{I}$ and a (goal) state $s \in S$ there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr_t^{\mathcal{D}}(\Diamond s) = \sup_{\mathcal{D}_i \models_U \mathcal{I}} Pr_t^{\mathcal{D}_i}(\Diamond s)$ for all transient states $t \in S_{tr}^{\mathcal{I}}$.

Proof 12
Consider any given enumeration $t_1, t_2, \ldots$ of $S^{\mathcal{I}}$. Clearly, this enumeration is finite, since $S^{\mathcal{I}}$ is finite, see assumption 1.

Construction. Now, let $\mathcal{D}^{(1)} \models_U \mathcal{I}$ with $Pr_{t_1}^{\mathcal{D}^{(1)}}(\Diamond s) = \overline{Pr}_{t_1}^{\mathcal{I}}$ and $T_1 := I_{t_1 s}^{\mathcal{D}^{(1)}}$, trivially such a DTMC must exist. By theorem 4.17, $Pr_u(\Diamond s) = \overline{\mathbf{P}}_u(\Diamond s)$.

Inductively, $\mathcal{D}^{(n)} \models_U \mathcal{I}$ (and $s \in T_n$). If $t_{n+1} \in T_n$, then we set $\mathcal{D}^{(n+1)} := \mathcal{D}^{(n)}$ and $T_{n+1} = T_n$. If not, we use lemma 4.19 with $T = T_n$, $\mathcal{D}_T = \mathcal{D}^{(n)}$, and $t = t_{n+1}$ to obtain $\mathcal{D}^{(n+1)} := \mathcal{D}_{cmb}$ with $\mathcal{D}^{(n+1)} \models_U \mathcal{I}$. We set $T_{n+1} = T_n \cup I_{t_{n+1} s}^{\mathcal{D}^{(n+1)}}$, thus and $Pr_{t_i}^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_{t_i}^{\mathcal{I}}(\Diamond s)$ for all $t_i \in T_{n+1}$, as $Pr_{t_i}^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_{t_i}^{\mathcal{I}}(\Diamond s)$ for all $t_i \in T_n$ (induction), $Pr_{t_{n+1}}^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_{t_{n+1}}^{\mathcal{I}}(\Diamond s)$ (see lemma 4.19), and $Pr_u^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t_{n+1} s}^{\mathcal{D}^{(n+1)}}$ (see theorem 4.17). □

Thus, to conclude, for any given IDTMC $\mathcal{I}$ and state s, there exists a $\mathcal{D}$ with optimal reachability probability for all states $t \in S^{\mathcal{I}}$

Remark. *Non-attainability of values in EVT intervals.* In general, not every value in $[\underline{\mathbb{E}}[vt_s], \overline{\mathbb{E}}[vt_s]]$ is attainable. In other words, there exists an IDTMC with state s, such that there exists a $x \in [\underline{\mathbb{E}}^{\mathcal{I}}[vt_s], \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]]$ with $\mathbb{E}^{\mathcal{D}}[vt_s] \neq x$ for all $\mathcal{D} \models \mathcal{I}$. This holds even for $\overline{\mathbb{E}}[vt_s] < \infty$.

We provide a short example for this remark.

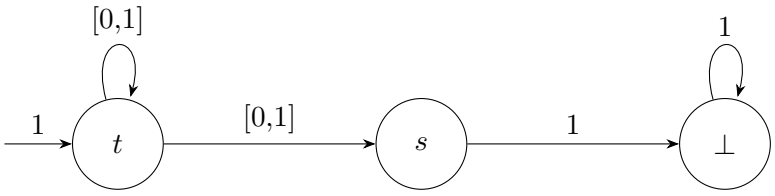


Figure 4.4: An example of a DTMC.

Example 4.21
Consider the IDTMC $\mathcal{I} = (\{t, s, \bot\}, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ shown in fig. 4.6. Clearly, $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 0$ and $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 1$. However, there exists no $\mathcal{D} \models_U \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}}[vt_s] = x$ for any $x \in (0, 1)$, as $\mathbb{E}^{\mathcal{D}}[vt_s] = 1$ if $\mathbf{P}^{\mathcal{D}}(t, s) > 0$ and $\mathbb{E}^{\mathcal{D}}[vt_s] = 0$ else.

4.2 Calculating Expected Visiting Times of IDTMCs

We will now have a look at how the *expected visiting times* for IDTMCs can be calculated

The most intuitive upper and lower bounds for the uEVTs and lEVTs of IDTMCs can be derived by using eq. (3.1) and the upper and lower transition probability functions and initial distributions defined in def. 4.1.

Theorem 4.22 (Upper and Lower Bounds for uEVTs and lEVTs)
For any IDTMC $\mathcal{I}$, the uEVT is bounded from above by $\overline{\mathbb{E}}[vt_s] \leq \mathbb{E}_{upp}[vt_s] = \overline{\iota}_{init}(s) + \sum_{t \in S_{tr}} \overline{\mathbf{P}}(t, s)^{\mathsf{T}} \cdot \mathbb{E}_{upp}[vt_t]$ and the ℓEVT is bounded from below by $\underline{\mathbb{E}}[vt_s] \geq \mathbb{E}_{low}[vt_s] = \underline{\iota}_{init}(s) + \sum_{t \in S_{tr}} \underline{\mathbf{P}}(t, s)^{\mathsf{T}} \cdot \mathbb{E}_{low}[vt_t]$ for all $s \in S^{\mathcal{I}}_{tr}$.

Proof 13
For any IDTMC $\mathcal{I}$ there exists a DTMC $\mathcal{D}$ that satisfies $\mathcal{I}$, i.e., $\mathcal{D} \models \mathcal{I}$, and $\mathbb{E}^{\mathcal{D}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$. As $\mathcal{D} \models \mathcal{I}$, it follows that $\mathbf{P}(s, s') \leq \overline{\mathbf{P}}(s, s')$ for all $s, s' \in S$.

Thus, $\mathbb{E}^{\mathcal{D}}[vt_s] = \iota_{init}(s) + \sum_{t \in S_{tr}} \mathbf{P}(t, s)^{\mathsf{T}} \cdot \mathbb{E}[vt_t] \leq \overline{\iota}_{init}(s) + \sum_{t \in S_{tr}} \overline{\mathbf{P}}(t, s)^{\mathsf{T}} \cdot \mathbb{E}_{upp}[vt_t] = \mathbb{E}^{\mathcal{I}}_{upp}[vt_s]$.

The identical argument can be made to prove the lower bound for the ℓEVT. □

As we fix the transition probability functions to either $\underline{\mathbf{P}}$ or $\overline{\mathbf{P}}$, we remove the uncertainty of the IDTMCs and thus obtain a DTMC with a transition matrix that is not row-stochastic. Therefore, we can calculate these bounds by using eq. (3.1) in PTIME. Moreover, as the transition matrix corresponding to $\underline{\mathbf{P}}$ is row-substochastic, and thus convergent as shown in [KS76], we can use value iteration to compute the

lower bounds $\mathbb{E}_{low}$.

However, while these bounds can be computed efficiently, they are not very tight.

Example 4.23
For this take a look at the upper bound of the IDTMC provided in fig. 4.5. Using theorem 4.22, we can calculate that $\mathbb{E}_{upp}[vt_{s_4}] = 2$, as $\mathbb{E}_{upp}[vt_{s_1}] = \mathbb{E}_{upp}[vt_{s_2}] = \mathbb{E}_{upp}[vt_{s_3}] = 1$. However, as the IDTMC is acyclic, see def. 3.17, every state can be visited at most once. Thus, the upper bound $\mathbb{E}_{upp}[vt_{s_4}]$ is clearly not tight. Similarly, $\mathbb{E}_{low}[vt_{s_4}] = 0$, however s_4 needs to be visited at least once. Thus, this lower bound $\mathbb{E}_{low}[vt_{s_4}]$ is not tight either.

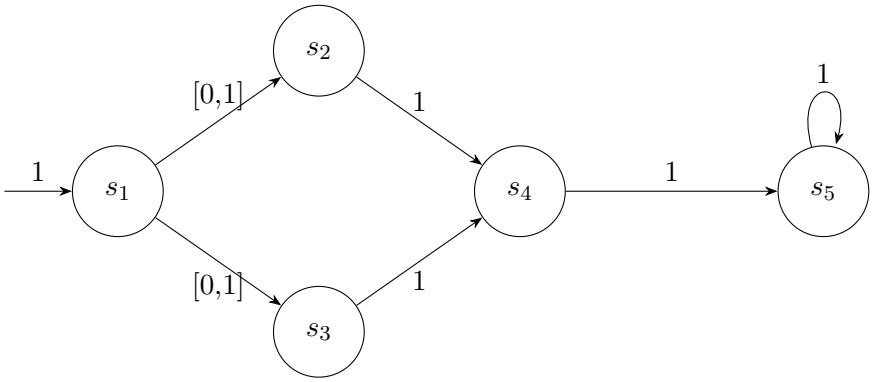


Figure 4.5: An example of a DTMC.

In the following, we will therefore take a look at how we can characterize the precise values of the uEVTs and lEVTs.

Theorem 4.24
For a given state s in an IDTMC $\mathcal{I}$, $\overline{\mathbb{E}}[vt_s] = \infty$ if and only if $s \in S^{\mathcal{I}}_{re}$.

Proof 14
By our assumption 1, there exists a DTMC $\mathcal{D} \models \mathcal{I}$ such that s is reachable in $\mathcal{D}$, i.e., $Pr^{\mathcal{D}}(\Diamond s) > 0$ and $\overline{Pr}^{\mathcal{I}}(\Diamond s) > 0$. Moreover, as $s \in S^{\mathcal{I}}_{re}$, there exists a DTMC $\mathcal{D}_{re} \models \mathcal{I}$ such that $Pr^{\mathcal{D}_{re}}_s(\bigcirc\Diamond s) = 1$, see def. 4.12, thus $\overline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s) = 1$. Using theorem 4.20, there exists a DTMC $\mathcal{D}_{+re} \models \mathcal{I}$ such that $Pr^{\mathcal{D}_{+re}}(\Diamond s) > 0$ and $Pr^{\mathcal{D}_{+re}}_{N(s)}(\Diamond s) = 1$, as $\overline{Pr}^{\mathcal{I}}(\Diamond s) > 0$ and $\overline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s) = 1$. Therefore, clearly $s \in S^{\mathcal{D}_{+re}}_{re}$, see def. 3.14 and using lemma 3.24, we can thus conclude that $\mathbb{E}^{\mathcal{D}_{+re}}[vt_s] = \infty$ and thus $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \infty$.

Conversely, if $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \infty$, then there exists a DTMC $\mathcal{D} \models \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}}[vt_s] = \infty$, i.e. $s \in S^{\mathcal{D}}_{re}$, see lemma 3.24. Thus, it follows directly that $s \in S^{\mathcal{I}}_{re}$, see def. 4.12. □

Theorem 4.25
For any $s \in S^{\mathcal{I}}$ in a given IDTMC $\mathcal{I}$, $\underline{\mathbb{E}}[vt_s] = 0$ if and only if $\underline{Pr}^{\mathcal{I}}(\Diamond s) = 0$, i.e., there exists a satisfying DTMC of $\mathcal{I}$, such that s is not reachable.

Proof 15
If $\underline{Pr}^{\mathcal{I}}(\Diamond s) = 0$, then there exists $\mathcal{D} \models \mathcal{I}$ such that $Pr^{\mathcal{D}}(\Diamond s) = 0$, i.e., there exists no path in $\mathcal{D}$ that contains s. As therefore, $vt_s(\pi) = 0$ for all $\pi \in Paths^{\mathcal{D}}_{inf}$, we can conclude, using def. 3.23, that $\mathbb{E}^{\mathcal{D}}[vt_s] = 0$ and thus $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 0$.

Conversely, if $\underline{Pr}^{\mathcal{I}}(\Diamond s) > 0$, then there exists $\pi \in Paths^{\mathcal{D}}_{inf}$ with $vt_s(\pi) > 0$ for all $\mathcal{D} \models \mathcal{I}$. Therefore, $\mathbb{E}^{\mathcal{D}}[vt_s] > 0$ and thus $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] > 0$. □

As we showed in theorem 4.20 in sec. 4.1.1, there exists a DTMC $\mathcal{D}_{max} \models_U \mathcal{I}$ for any IDTMC $\mathcal{I}$ with $Pr^{\mathcal{D}}_t(\Diamond s) = \overline{Pr}^{\mathcal{I}}_t(\Diamond s)$ for all $t \in S^{\mathcal{D}}$, see theorem 4.20, we can conclude that $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \mathbb{E}^{\mathcal{D}_{max}}[vt_s]$ under the UMC semantic, using eq. (3.2). In other words, we can calculate the uEVT of a state s in a given IDTMC by determining the maximum reachability probability of $Pr^{\mathcal{I}}_t(\Diamond s)$ for all $t \in S_{tr \cap in}$ and $Pr_s(\bigcirc \Diamond s)$.

As shown in [CHK13], computing the maximum or minimum reachability probability is P-complete for the IMDP semantic, and thus also for the UMC semantic.

We will now shortly show that computing ℓEVTs and uEVTs is P-complete, by reducing it to the computation of minimum and maximum reachability probabilities, which is P-complete.

Theorem 4.26
The problem of uEVTs and ℓEVTs is P-complete.

Reduction. Given an IDTMC $\mathcal{I}$ and a set of states $T \subseteq S$, we construct the IDTMC $\mathcal{I}_{red} = (S^{\mathcal{I}} \cup \{s, \bot\}, \mathbf{P}^{\mathcal{I}_{red}}, \iota^{\mathcal{I}}_{init})$. We define $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u, w) = \overline{\mathbf{P}}^{\mathcal{I}}(u, w)$ and $\underline{\mathbf{P}}^{\mathcal{I}_{red}}(u, w) = \underline{\mathbf{P}}^{\mathcal{I}}(u, w)$ for all $u \in S^{\mathcal{I}} \backslash T$ and $w \in S^{\mathcal{I}}$. Moreover, $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u, w) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(u, w) = 0$ and $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u, s) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(u, s) = 1$ for $u \in T$ and $w \in S^{\mathcal{I}}$. Finally, $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(s, \bot) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(s, \bot) = 1$ and $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(\bot, \bot) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(\bot, \bot) = 1$.

Lemma 4.27
The mapping $(\mathcal{I}, T) \mapsto (\mathcal{I}_{red})$ is computable in logarithmic space.

Theorem 4.28
Let $\mathcal{D}_{red} \models_U \mathcal{I}_{red}$ denote a DTMC such that $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$ and $\mathcal{D}_{max} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}_{max}}(\Diamond T) = \overline{Pr}^{\mathcal{I}}(\Diamond T)$. Then, $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = Pr^{\mathcal{D}_{max}}(\Diamond T)$

Lemma 4.29
$\overline{Pr}^{\mathcal{I}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$ and $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond s)$

Proof 16
As $\mathbf{P}^{\mathcal{D}}(t,s) > 0$ if and only if $t \in T$ for all $\mathcal{D} \models_U \mathcal{I}_{red}$, it follows from lemma 3.11 that $Pr^{\mathcal{D}}(\Diamond s) = \sum_{t \in T} \mathbf{P}(t,s) \cdot Pr^{\mathcal{D}}(\Diamond T)$. Because, $\mathbf{P}^{\mathcal{D}}(t,s) = 1$ for all $t \in T$, we can conclude that $Pr^{\mathcal{D}}(\Diamond T) = Pr^{\mathcal{D}}(\Diamond s)$ for all $\mathcal{D} \models_U \mathcal{I}_{red}$. □

Lemma 4.30
$\mathbb{E}^{\mathcal{I}_{red}}[vt_s] = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$ and $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = Pr^{\mathcal{D}_{red}}(\Diamond s)$

Proof 17
$\mathbb{E}^{\mathcal{D}}[vt_s] = Pr(\Diamond s) \cdot (F)_{ss}$ for all $\mathcal{D} \models_U \mathcal{I}_{red}$, see lemma 3.27. As, $\overline{Pr}^{\mathcal{I}_{red}}(s,t) > 0$ if and only if $t = \tau$ and τ is absorbing it follows that $(F)_{ss} = 1$ which concludes the proof. □

Lemma 4.31
$Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{max}}(\Diamond T)$

Proof 18
Firstly, if $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$, then $Pr^{\mathcal{D}_{red}}(\Diamond s) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$, see lemma 4.30. Moreover, $Pr^{\mathcal{D}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond T)$, see lemma 4.29.

As $Pr^{\mathcal{D}_{red}}(\Diamond T)$ and $Pr^{\mathcal{D}_{max}}(\Diamond T)$ are only dependent on $\mathbf{P}^{\mathcal{D}_{red}}(u,w)$ and $\mathbf{P}^{\mathcal{D}}(u,w)$ with $u \in S^{\mathcal{I}} \backslash T$ respectively, see lemma 3.11, and $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u,w) = \overline{\mathbf{P}}^{\mathcal{I}}(u,w)$ as well as $\underline{\mathbf{P}}^{\mathcal{I}_{red}}(u,w) = \underline{\mathbf{P}}^{\mathcal{I}}(u,w)$ for all $u \in S^{\mathcal{I}} \backslash T$ and $w \in S^{\mathcal{I}}$, thus trivially $\overline{Pr}^{\mathcal{I}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond T)$ which concludes the proof. □

We can now prove theorem 4.28.

Proof 19
We simply need to combine the results of the lemmata. According to lemma 4.31 $Pr^{\mathcal{D}_{max}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond T)$. Furthermore, using lemma 4.29, $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond s)$. Finally, lemma 4.30 shows that $Pr^{\mathcal{D}_{red}}(\Diamond s) = \mathbb{E}^{\mathcal{D}_{red}}[vt_s]$. Thus, com-

bining these results $Pr^{\mathcal{D}_{max}}(\Diamond T) = \mathbb{E}^{\mathcal{D}_{red}}[vt_s]$. By definition $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$ and $Pr^{\mathcal{D}_{max}}(\Diamond T) = \overline{Pr}^{\mathcal{I}}(\Diamond T)$. Therefore, the given maximum reachability probability $\overline{Pr}^{\mathcal{I}}(\Diamond T)$ of any IDTMC $\mathcal{I}$ can be determined through calculating uEVTs.□

Remark. Konstruktion: jeweils zusammengefasste init states und direct successors von s. −¿ zeigen, dass das die selben EVTs hat (Bijektion für DTMCs). Zeigen das IDTMC Konstruktion nach lemma DTMC mit max EVT hat (beide max reach probs) (vllt reicht es auch zu zeigen dass es für diesen DTMC ein Äquivalent im ursprünglichen IDTMC gibt das selbe EVTs hat (für s halt)) eq. (3.3) $\mathbb{E}^{\mathcal{D}}[vt_s] = \frac{Pr_{s_{in}}^{\mathcal{D}_{uni}}(\Diamond s)}{1-Pr_{s_{suc}}^{\mathcal{D}_{uni}}(\Diamond s)} \cdot \tau_{init}(s_{in})$

\+ Zwei state Konstruktion schnelle Möglichkeit zugrunde liegende Struktur zu berechnen (einfach geaddete states in transitionen übersetzen)

UMC & IMDP selbe max/min EVTs

→ Bsp zum verdeutlichen

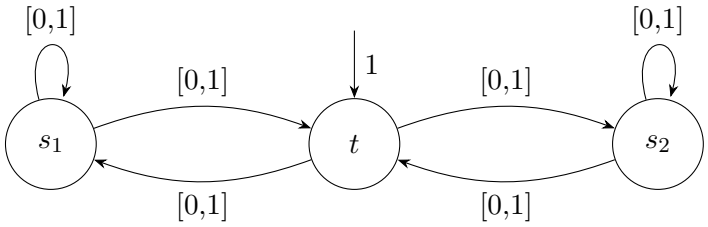


Figure 4.6: An example of a DTMC.

4.2.1 UMC semantic

In this section we examine how the precise values of the uEVTs and lEVTs can be calculated for IDTMCs under the UMC semantic.

As defined, there can be multiple DTMCs $\mathcal{D}_i$ that satisfy an IDTMC $\mathcal{I}$ with differing transition probability functions $\mathbf{P}^{\mathcal{D}_i}$ and initial distributions $\iota_{init}^{\mathcal{D}_i}$. To determine the uEVT and ℓEVT for a given state $s \in S^{\mathcal{I}}$, we thus have to find the satisfying DTMCs that maximize and minimze eq. (3.1), respectively. In other words, we need to solve the equations

$$\mathbb{E}[vt_s] = \text{opt}_{\mathcal{D} \models \mathcal{I}} \left(\iota_{init}(s) + \sum_{t \in S_{tr}} \mathbf{P}^{\mathcal{D}}(t,s) \cdot \mathbb{E}[vt_t] \right), \qquad \text{opt} \in \{\min, \max\}.$$

As we concluded, the ℓEVT and uEVT of a given state s in an IDTMC, can be calculated by computing the reachability probabilities $\overline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\overline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$ or

$\underline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\underline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$. In [CHK13], the authors construct the following NLP to compute any maximum or minimum reachability, respectively, for a set of nodes $T \subseteq S$, using the fixed-initial-state-definition.

$$
\begin{aligned}
&\text{opt} \quad \sum_{s\in S\setminus T} x_s && \text{opt} \in \{\max, \min\} && \\
&\text{s.t.} \quad x_s \sim \sum_{s'\in S\setminus T} \mathbf{P}(s,s')x_{s'} + \sum_{s'\in T} \mathbf{P}(s,s') && \forall s \in S\setminus T \text{ and } \sim\in\{\leq,\geq\} && (1)\\
&\qquad x_s = 1 && \forall s \in T && (2)\\
&\qquad \sum_{s_j\in N(s)} \mathbf{P}(s_i,s_j) = 1 && \forall s_i \in S_{tr} && (3)\\
&\qquad \underline{\mathbf{P}}^{\mathcal{I}}(s_i,s_j) \leq \mathbf{P}(s_i,s_j) \leq \overline{\mathbf{P}}^{\mathcal{I}}(s_i,s_j) && \forall s_i,s_j \in S^{\mathcal{I}}_{tr} && (4)
\end{aligned}
$$

(Allowed only max + $\leq$ and min + $\geq$)

4.2.2 IMDP semantic

In this subsection we will look at how uEVTs and lEVTs can be calculated for the IMDP semantic.

Recall, that we concluded in sec. 4.1.1 that the uEVT and ℓEVT of a given IDTMC $\mathcal{I}$ can be calculated by maximizing or minimizing the term $\sum_{t\in S_{tr}\cap in} \frac{Pr_t(\Diamond s)}{1-Pr_s(\bigcirc\Diamond s)} \cdot \iota_{init}(t)$ for all DTMCs $\mathcal{D} \models \mathcal{I}$. Intuitively, we can thus find the lEVTs and uEVTs by maximizing or minimizing the reachability probability from the initial state t to the given state s ($Pr_t(\Diamond s)$) and from s back to itself ($Pr_s(\bigcirc\Diamond s)$). As shown in [CHK13] the problem of finding this supremum or infimum for a given IDTMC is P-complete. Thus, we can derive the two suprema or infima belonging to (at most) two different satisfying DTMCs $\mathcal{D}_{s_{in}\to s}$ and $\mathcal{D}_{s\to s}$ ($\mathcal{D}_{s_{in}\to s},\ \mathcal{D}_{s\to s} \models_I \mathcal{I}$).

As we have defined the IMDP semantic using adversarial behaviour (?), notice that we can define a policy that choses the probability distribution specified by $\mathcal{D}_{s_{in}\to s}$ until state s is visited for the first time and then choses the probability distribution specified by $\mathcal{D}_{s\to s}$. This policy can be characterized as having two stages where we transition from the first to the second stage when passing through s for the first time. Notice however, that the stochastic process of the Markov Chain/MDP (?) would no longer be memoryless, i.e. the Markov Property (SEE ...) would break, as the process is not only dependent on the current state but also on what stage the policy is in. To solve this problem, we propose a remapping/reduction (state augmentation).

Construction. Given an IDTMC $\mathcal{I} = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$, the previously defined $\mathcal{D}_{s_{in}\to s}, \mathcal{D}_{s\to s}$ (corresponding to either the suprema or infima) and a state s (for which we want to calculate the EVT), we define $\mathcal{D}^s_{aug}$, with $S^{\mathcal{D}^s_{aug}} = S_{st1} \cup S_{st2}$, where the state set for stage 1 is $S_{st1} = \{s^{(1)} \mid s \in S^{\mathcal{I}}_{tr}\}$ (transient states in $\mathcal{I}$) and the state set for

stage 2 is $S_{st2} = \{s^{(2)} \mid s \in S^{\mathcal{I}}\}$. Furthermore, $\mathbf{P}^{\mathcal{D}^s_{aug}}(s_1^{(1)}, s_2^{(1)}) = \mathbf{P}^{\mathcal{D}_{s_{in}\to s}}(s_1, s_2)$ for all $s_1^{(1)} \in S_{st1}$, $s_2^{(1)} \in S_{st1}\backslash\{s\}$ and $\mathbf{P}^{\mathcal{D}^s_{aug}}(s_1^{(1)}, s_2^{(2)}) = \mathbf{P}^{\mathcal{D}_{s_{in}\to s}}(s_1, s_2)$ for $s_1^{(1)} \in S_{st1}$, $s_2^{(2)} \in S_{st2} \cap S^{\mathcal{I}}_{re}$. Similarily, $\mathbf{P}^{\mathcal{D}^s_{aug}}(s_1^{(2)}, s_2^{(2)}) = \mathbf{P}^{\mathcal{D}_{s\to s}}(s_1, s_2)$ for all $s_1^{(2)}, s_2^{(2)} \in S_{st2}$. Moreover, $\mathbf{P}^{\mathcal{D}^s_{aug}}(s^{(1)}, s_i^{(2)}) = \mathbf{P}^{\mathcal{D}_{s\to s}}(s, s_i)$ for $s^{(1)} \in S_{st1}$ and $s_i \in N^{\mathcal{I}}(s)$. Lastly, $\iota^{\mathcal{D}^s_{aug}}_{init}(t^{(1)}) = \iota^{\mathcal{D}_{s_{in}\to s}}_{init}(t)$ for all $t^{(1)} \in S_{st1}$ and $\iota^{\mathcal{D}^s_{aug}}_{init}(t^{(2)}) = \iota^{\mathcal{D}_{s_{in}\to s}}_{init}(t)$ for all $t^{(2)} \in S_{st2} \cap S_{re}$, while $\iota^{\mathcal{D}^s_{aug}}_{init}(t^{(2)}) = 0$ for all $t^{(2)} \in S_{st2}\backslash S_{re}$.

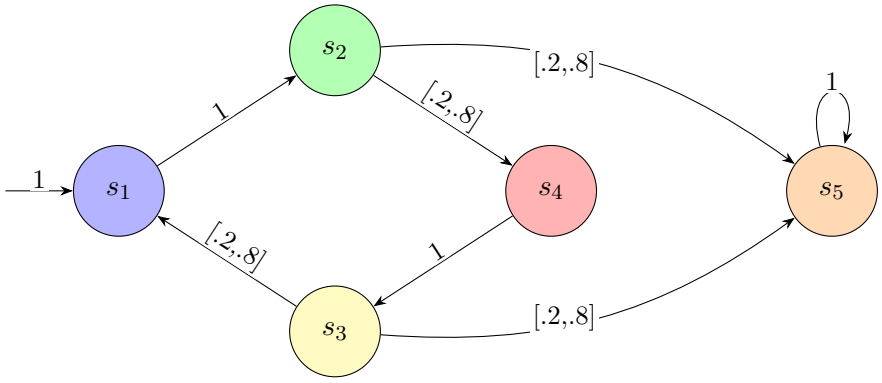


Figure 4.7: target state in red

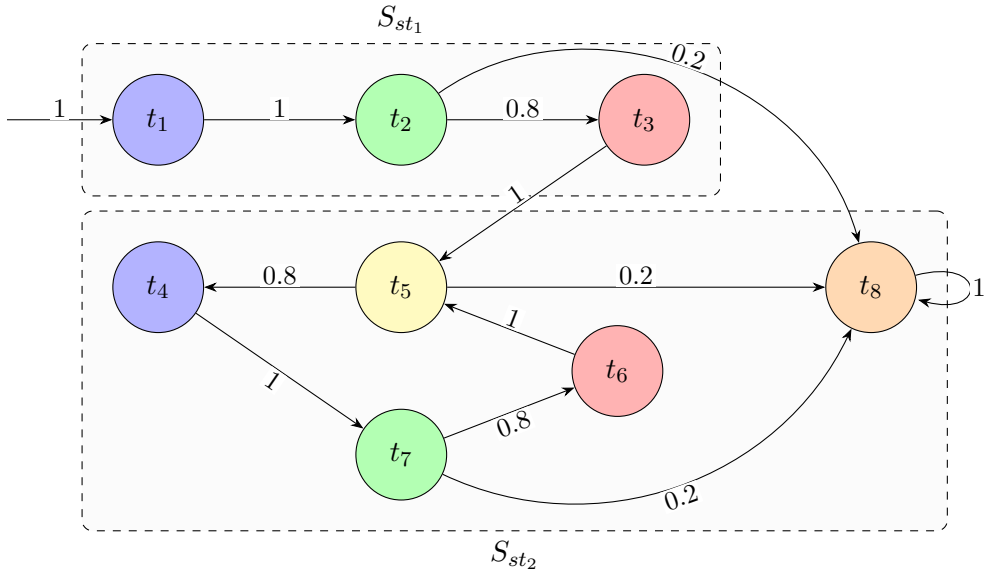


Figure 4.8: $\tau(t_1) = \tau(t_4) = s_1$, $\tau(t_2) = \tau(t_7) = s_2$, $\tau(t_5) = s_3$, $\tau(t_3) = \tau(t_6) = s_4$. Target state in red.

Example 4.32
Recall the IDTMC $\mathcal{I}$ in fig. 4.1, displayed again in fig. 4.7. fig. 4.8 shows $\mathcal{D}^{s_4}_{aug}$, i.e., the construction of the maximizing DTMC for the EVT of $\mathcal{I}$.

(We will now show that this remapping / characterization of this history-dependent policy is coincides (?) with def. 4.5.)

Lemma 4.33
Given and IDTMC $\mathcal{I}$ and the corresponding $\mathcal{D}^s_{aug}$, it holds that $\mathcal{D}^s_{aug} \models_I \mathcal{I}$

Proof 20
To prove this lemma, we will show that def. 4.5 holds for $\mathcal{D}^s_{aug}$. For this, we will first define a mapping $\tau : S^{\mathcal{D}^s_{aug}} \to S^{\mathcal{I}}$. For this, we need to account for S_{st1} and S_{st2}. We define $\tau(s_i^{(1)}) = s_i$ for all $s_i^{(1)} \in S_{st1} \cap S_{tr}$ and $\tau(s_i^{(2)}) = s$ for all $s_i^{(2)} \in S_{st2}$. Thus, for all $s \in S^{\mathcal{I}}_{tr}$, there exists precisely one $s^{(1)} \in S_{st1}$ with $\tau(s^{(1)}) = s$ and $\iota^{\mathcal{D}_{aug}}_{init}(s^{(1)}) \in [\underline{\iota}^{\mathcal{I}}_{init}(s), \overline{\iota}^{\mathcal{I}}_{init}(s)]$ and $s^{(2)} \in S_{st2}$ with $\tau(s^{(2)}) = 0$. Moreover, for all $s \in S^{\mathcal{I}}_{re}$, there exists precisely one $s^{(2)} \in S_{st2}$ with $\tau(s^{(2)}) = s$ and $\iota^{\mathcal{D}_{aug}}_{init}(s^{(2)}) \in [\underline{\iota}^{\mathcal{I}}_{init}(s), \overline{\iota}^{\mathcal{I}}_{init}(s)]$ and $s^{(1)} \notin S_{st1}$. Therefore, we can conclude that condition (i) of def. 4.5 holds.

Secondly, condition (ii) clearly holds for all $s_i^{(2)} \in S_{st2}$, as all of its transition are defined to be identical to that of $\mathcal{D}_{s\to s}$ and $\mathcal{D}_{s\to s} \models \mathcal{I}$. Thus, it remains only to show that condition (ii) also holds for all states in S_{st1}. For all states $s_1^{(1)} \in S_{st1}\backslash\{s\}$, $\mathbf{P}^{\mathcal{D}^s_{aug}}(s_1^{(1)}, s_2^{(1)}) = \mathbf{P}^{\mathcal{D}_{s_{in}\to s}}(s_1, s_2)$, for all $s_2 \in S^{\mathcal{D}_{s_{in}\to s}}_{tr}$ and $s_2 \in S^{\mathcal{D}_{s_{in}\to s}}_{re}$. Moreover, for $\mathbf{P}^{\mathcal{D}^s_{aug}}(s^{(1)}, s_i^{(1)}) = 0$ and $\mathbf{P}^{\mathcal{D}^s_{aug}}(s^{(1)}, s_i^{(2)}) = \mathbf{P}^{\mathcal{D}_{s\to s}}(s, s_i)$. As, $\mathcal{D}_{s_{in}\to s}$, $\mathcal{D}_{s\to s} \models \mathcal{I}$, we can thus conclude that condition (ii) also holds for all states in S_{st1}. As both conditions of def. 4.5 are fulfilled, we can thus conclude that $\mathcal{D}^s_{aug} \models_I \mathcal{I}$. □

Construction. For a given DTMC $\mathcal{D}$ and a state $s \in S$, consider the DTMC $\mathcal{D}_{abs} = (S^{\mathcal{D}_{abs}}, \mathbf{P}^{\mathcal{D}_{abs}}, \iota^{\mathcal{D}}_{init})$ with the set of states $S^{\mathcal{D}_{abs}} = S^{\mathcal{D}}_{tr} \cup \{s_{re}\}$. Moreover, the probability function $\mathbf{P}_{abs}$ is identical to $\mathbf{P}^{\mathcal{D}}$, except the transitions $\mathbf{P}_{abs}(s, s') = 0$ for all $s' \in S$ and $\mathbf{P}_{abs}(s, s) = 1$, as well as $\mathbf{P}^{\mathcal{D}}(s_1, s_2) = \mathbf{P}^{\mathcal{D}_{abs}}(s_1, s_{re})$, for all $s_1 \in S^{\mathcal{D}}_{tr}$, $s_2 \in S^{\mathcal{D}}_{re}$ and further $\mathbf{P}^{\mathcal{D}_{abs}}(s_{re}, s_{re}) = 1$. In other words, $\mathcal{D}_{abs}$ is modified only in such a way that s is absorbing and all recurrent states are summarized into one state.

Lemma 4.34
The reachability probabilities $Pr^{\mathcal{D}}(\Diamond s) = Pr^{\mathcal{D}_{abs}}(\Diamond s)$ are equal.

Proof 21
If the set of initial states S_{init} includes s then trivially the reachability probability is 1 for both $\mathcal{D}$ and $\mathcal{D}_{abs}$.

As $Pr(\Diamond s) = Pr(S\backslash\{s\} \cup s)$ (until operator), we can reason/conclude using def. 3.5 and def. 3.8 that $Pr(\Diamond s) = \iota^{\mathcal{D}}_{init}(\pi[0]) \cdot \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1])$. As there is no factor $\mathbf{P}(s, s')$ for any $s' \in S$ in this product, as well as no factor of the form $\mathbf{P}(s_1, s_2)$, where $s_1 \in S^{\mathcal{D}}_{re}$ and the probabilty functions $\mathbf{P}$ and $\mathbf{P}_{abs}$ are identical for all transitions except those, we can conclude that lemma 4.34 is true. □

Thus, we can reason that transitions from the state s itself and recurrent states have no influence on the reachability probability $Pr^{\mathcal{D}}(\Diamond s)$.

Lemma 4.35
Let $Pr^{max}(\Diamond s)$ and $Pr^{max}_s(\bigcirc\Diamond s)$ be the maximum reachability probabilities of $\mathcal{I}$. It holds that $Pr^{D^s_{aug}}(\Diamond s) = Pr^{max}(\Diamond s)$ and $Pr_s^{D^s_{aug}}(\bigcirc\Diamond s) = Pr^{max}_s(\bigcirc\Diamond s)$

Proof 22
First, we will show that $Pr^{D^s_{aug}}(\Diamond s) = Pr^{max}(\Diamond s)$. Notice, that for all $s \in S_{st1}\backslash\{s\}$, the probability function $\mathbf{P}^{\mathcal{D}^s_{aug}}$ is identical to $\mathbf{P}^{\mathcal{I}}$, i.e. it is identical for all transient states except s. Using lemma 4.34, we can thus conclude that $Pr^{D^s_{aug}}(\Diamond s) = Pr^{max}(\Diamond s)$.

Next, we will show that $Pr_s^{D^s_{aug}}(\bigcirc\Diamond s) = Pr^{max}_s(\bigcirc\Diamond s)$. Trivial because (basicaly/literaly) the same DTMC. □

4.2.3 Special Case: Acyclic IDTMCs

Definition 4.36 (Acyclic IDTMC)
An IDTMC $\mathcal{I}$ is acyclic if and only if all satisfying DTMCs $\mathcal{D}$ ($\mathcal{D} \models \mathcal{I}$) are acyclic.

– l/uEVT is simply maximum/minimum reachability (which can be calc in P, cite paper).

4.2.4 Value Iteration

Explain why useful & interesting to define Bellman operator usw. (Runtime advantages, etc.)

In [KTDB19], the authors prove

4.3 Exkurs: aIMCs

Definition 4.37 (Augmented Interval Markov Chains)
An augmented Interval Markov Chain (aIMC) $\mathcal{A}$ is a tuple $(S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init}, C)$, where $(S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ is an IDTMC $\mathcal{I}$ and $C \subseteq S^4$ is a set of *edge equality constraints*. Similar to the once-and-for-all semantic for IDTMCs, see **??**, we define, that an arbitrary DTMC $\mathcal{D}$ satisfies an aIMC $\mathcal{A}$ if it satisfies the corresponding IDTMC $\mathcal{I}$ and $\mathbf{P}^{\mathcal{D}}(s_1, s_2) = \mathbf{P}^{\mathcal{D}}(s_3, s_4)$ for all $(s_1, s_2, s_3, s_4) \in C$. [Cho17]

In other words, aIMCs extend the definition of IMCs with an "edge-constraint" which enforces certain transition probabilities to be equal. A restriction of the quantitative reachability problem, which we discussed in sec. 3.1.1 and sec. 4.1.1, is the qualitative reachability problem, where is restricted to $p \in \{0, 1\}$ (see sec. 4.1.1).

In [Cho17], the authors prove that the qualtitative reachability problem for aIMCs is NP-complete. Thus, as trivially the qualitative problem can be reduced to the quantitative problem, we can thus deduce that the quantitative problem for aIMCs is NP-hard. As we have shown in ...

4.4 Stationary Distributions in IDTMCs

As stated in theorem 3.3 the k-th power of the transition matrik $\mathcal{T}^k$ contains the probabilities of transitioning from any state i to any other state j in precisely k steps. Thus, the product $\iota_{init}^{\mathsf{T}} \cdot \mathcal{T}^k$ characterizes the probability distribution of the current state of the given DTMC in step k. We define this product as $\theta_k^{\mathsf{T}} = \iota_{init}^{\mathsf{T}} \cdot \mathcal{T}^k$, where the i-th entry of θ_k characterizes the probability that the current state in step k is i. We can also define θ_k iteratively through $\theta_k^{\mathsf{T}} = \theta_{k-1}^{\mathsf{T}} \cdot \mathcal{T}$

Definition 4.38 (Stationary Distributions in DTMCs)
A distribution $\theta_k^{\mathcal{D}}$ in a DTMC $\mathcal{D}$ is stationary if and only if $\theta_k^{\mathsf{T}} = \theta_k^{\mathsf{T}} \cdot \mathcal{T}^{\mathcal{D}}$, i.e. the probability distribution of the current state remains unchanged through future steps. We denote the stationary distribution as $\theta^{\mathcal{D}}$, where $\theta^{\mathcal{D}}(s)$ denotes the entry of $\theta^{\mathcal{D}}$ which corresponds to state s.

Definition 4.39 (Stationary Distributions in IDTMCs)
For a given IDTMC $\mathcal{I}$, we define $\Theta = \{\theta^{\mathcal{D}} \mid \mathcal{D} \models \mathcal{I}\}$ as the set of stationary distributions. Furthermore, for a given state $s \in S^{\mathcal{I}}$, we define $\underline{\theta}^{\mathcal{I}}(s)$ and $\overline{\theta}^{\mathcal{I}}(s)$ such that $\underline{\theta}^{\mathcal{I}}(s) = \inf_{\mathcal{D} \models \mathcal{I}} \theta^{\mathcal{D}}(s)$ and $\overline{\theta}^{\mathcal{I}}(s) = \sup_{\mathcal{D} \models \mathcal{I}} \theta^{\mathcal{D}}(s)$.

In contrast to EVTs, when considering the stationary distribution of a DTMC or IDTMC, we are interested in the recurrent, not transient, states.

Theorem 4.40
If θ is stationary, then $\theta(i) = 0$ if the corresponding state s_i is transient and $\theta(i) > 0$ if s_i is recurrent.

We provide a short proof.

Proof 23
Assume that some s is transient and for the stationary distribution $\theta(s) > 0$. As θ is stationary, there exists some $k \in \mathbb{N}$ such that $\theta_{k'}$ is stationary for $k' \geq k$. Thus, there is a nonzero probability that the current state in step $k' \geq k$ is s. However, as s is transient it follows that $Pr_s(\bigcirc\Diamond s) < 1$, see def. 3.14. Therefore, there must be some $k'' > k$ such that $\theta_{k''}(i) = 0$. This contradicts the stationary property of θ. Now, assume that some s is recurrent and for the stationary distribution $\theta(s) = 0$. As s is reachable by definition/construction, see **??**, we can conclude that $Pr(\Diamond s) > 0$. Thus, there exists some $k' \in \mathbb{N}$ such that $\theta_{k'}(i) > 0$. As s_i is recurrent, it follows that $Pr_{s_i}(\bigcirc\Diamond s_i) = 1$, see again def. 3.14. Thus, there exist infinitely many $k'' \in \mathbb{N}$ with $k'' > k'$ and $\theta_{k''}(i) > 0$. Therefore, θ_k is not stationary. □

Notice, that any maximal set of connected recurrent states, i.e., every state is connected to some other state in the set, in a DTMC, is a BSCC, see def. 3.13. Firstly, a set of connected recurrent states in a DTMC is strongly connected. By definition, see **??**, all of the recurrent states in the set are reachable. As the return probability $Pr_{s_{re}}(\bigcirc\Diamond S_{re})$ is equal to 1 for any recurrent state s_re, see def. 3.14, any state that is a successor to s_{re} must also be a predecessor, i.e., every edge in this set is bidirectional (in a sense). Thus, trivially strongly connected.

→ It is SCC

Clearly also BSCC

(Bridge: recurrent groups are BSCC & BSCC is irreducible DTMC (perhaps prove this shortly)) We denote stationary distribution of a DTMC $\mathcal{D}$ restricted to a BSCC B as $\theta^{\mathcal{D}|_B}$.

As proven in [Kul16], the stationary distribution $\theta^{\mathcal{D}}$ of the recurrent states of a DTMC $\mathcal{D}$ is characterized by

$$\theta^{\mathcal{D}}(s) = Pr^{\mathcal{D}}(\Diamond B) \cdot \theta^{\mathcal{D}|_B}(s) \ , \tag{4.2}$$

for all $s \in B$. Moreover, in [MKQW25], it is shown that the stationary distribution within a BSCC can be computed through its EVTs. For this they construct a DTMC $\mathcal{B}\,\hat{\tau}^{\hat{v}}$ for every BSCC B with any arbitrary state $v \in S^{\mathcal{D}|_B}$ in which all incoming transitions of v are redirected to the fresh absorbing state $\hat{v}$.

Construction. More precisely, $\mathcal{B}\,\hat{\tau}^{\hat{v}} = (S^{\mathcal{D}|_B} \cup \{\hat{v}\}, \mathbf{P}^{\mathcal{B}\hat{\tau}^{\hat{v}}}, \iota_{init}^{\mathcal{B}\hat{\tau}^{\hat{v}}})$, where $\mathbf{P}^{\mathcal{B}\hat{\tau}^{\hat{v}}}(\hat{v}, \hat{v}) = 1$, $\mathbf{P}^{\mathcal{B}\hat{\tau}^{\hat{v}}}(s, \hat{v}) = \mathbf{P}(s, v)$ and $\mathbf{P}^{\mathcal{B}\hat{\tau}^{\hat{v}}}(s, v) = 0$ for all $s \in S^{\mathcal{D}|_B}$, $\mathbf{P}^{\mathcal{B}\hat{\tau}^{\hat{v}}}(s, t) = \mathbf{P}(s, t)$ for all

$s \in S$, $t \in S \backslash \{v\}$, and $\iota_{init}^{\mathcal{B}^{\not\circlearrowright \hat{v}}}(v) = 1$.

The stationary distribution within a DTMC is then given by

$$\theta^{\mathcal{D}|_B}(s) = \frac{\mathbb{E}^{\mathcal{B}^{\not\circlearrowright \hat{v}}}[vt_s]}{\sum_{t \in S} \mathbb{E}^{\mathcal{B}^{\not\circlearrowright \hat{v}}}[vt_t]} \ , \tag{4.3}$$

as proven in [MKQW25].

Assumption 2
In the rest of this section, we assume that all IDTMCs are graph preserving, meaning that there exist no lower transition probability function such that $\underline{\mathbf{P}}^{\mathcal{I}}(s,t) = 0$ for any $s, t \in S^{\mathcal{I}}$.

Notice, that all DTMCs $\mathcal{D}_i$ (for UMC semantic) satisfying an IDTMC $\mathcal{D}_i \models \mathcal{I}$ have the same edge relations, i.e., if $\mathbf{P}^{\mathcal{D}_j}(u,w) > 0$ for some $u, w \in S^{\mathcal{I}}$, then $\mathbf{P}^{\mathcal{D}_k}(u,w) > 0$ for all $\mathcal{D}_k$. (schlechte notation; auch wieder: eig u,w in $S^{\mathcal{D}}$, bzw. die states in $\mathcal{D}$, die auf gewieses u in IMC verweisen).

As all IDTMCs are graph preserving, we can conclude that for a given IDTMC $\mathcal{I}$, any state $s \in S_{re}^{\mathcal{I}}$ and BSCC B, if $s \in B$ then $s \notin B'$ for all BSCC $B' \neq B$. In other words, every state $S_{re}^{\mathcal{I}}$ corresponds to precisely one BSCC in $\mathcal{I}$.

Proof 24
Assume that for some IDTMC $\mathcal{I}$ there exists a state s and two BSCCs $B^{\mathcal{D}}, B'^{\mathcal{D}'}$ with $B^{\mathcal{D}} \neq B'^{\mathcal{D}'}$ and $\mathcal{D}, \mathcal{D}' \models \mathcal{I}$ such that $s \in B$ and $s \in B'$. As $B \neq B'$, there exists some $t \in (B \backslash B') \cup (B' \backslash B)$, i.e., there exists some $t \in S^{\mathcal{I}}$ only in B or B'. Let us assume $t \in B^{\mathcal{D}}$. (ist das ok so, halt um triviale Fallunterscheidung zu vermeiden) Thus by definition, see def. 3.13, there exists a path from s to t in $\mathcal{D}$. However, as noted the graph relations for all satisfying DTMCs of graph preserving IDTMCs are identical. Thus, there must also exist a path from s to t in $\mathcal{D}'$. However, $t \notin B'^{\mathcal{D}'}$, which contradicts def. 3.13. □

Thus, we can conclude that for graph preserving IDTMCs that any state $s \in S^{\mathcal{I}}$ belongs only to a singular BSCC, i.e., the BSCCs for all satisfying DTMCs are identical.

Therefore, we can calculate the stationary distributions

$$\underline{\theta}^{\mathcal{I}}(s) = \underline{Pr}^{\mathcal{I}}(\Diamond s) \cdot \frac{\underline{\mathbb{E}}^{\mathcal{B}^{\not\circlearrowright \hat{v}}}[vt_s]}{\sum_{t \in S} \mathbb{E}^{\mathcal{B}^{\not\circlearrowright \hat{v}}}[vt_t]} \ , \tag{4.4}$$

similar to eq. (4.2) using eq. (4.3).

Show: when $\mathbb{E}[vt_s]$ minimal then sum of $\mathbb{E}[vt_t]$ is maximal (and other way around). Bzw die summe aller evts ist für alle satisfying DTMCs gleich (–¿ nicht wahr; vllt auch ein interessantes Ergebnis??)

This directly concludes the optimality of the BSCC restricted stat. distribution.

(Trivially then uppper bound for stationary distribution given by uEVT and lower bound given by ℓEVT)

Chapter 5

Main Contribution

Chapter 6

Conclusion

Appendix A

Appendix

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